Supplement to "Who Is Really on the Frontier? Confidence Sets for Cost-Accuracy Pareto Leaderboards"

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Every table here is generated from the result files by `src/make_supplement.py`; the numbers are the ones the main text uses. System names are the agent-and-model part of HAL's run names; a dagger marks a board whose costs are run totals spread evenly over the tasks and treated as fixed.

S1. Every system on every board

Cost per task is in dollars under one price table with each model call counted once. "Front." marks the observed frontier, "Set" the frontier confidence set at the 5% level, "Fixed" the same set with costs fixed at their means, "Rob." the tie-robust set, "Stab." the bootstrap frontier stability, and "Best" the confidence set of the most accurate system. The rival lead ("Lead") is the strongest cheaper-or-equal system's accuracy minus this system's, in paired standard errors; a system leaves the frontier set when its rival lead exceeds the board's simultaneous margin.

Table S1: AssistantBench: 32 systems on 33 tasks, cheapest first. Unc.: tasks with an outcome but no logged model call, priced at the run's mean costed task. Front.: on the observed frontier. Set: in the frontier set. Fixed: in the frontier set with every cost fixed at its mean. Rob.: in the tie-robust set. Stab.: share of task resamples in which the system is on the observed frontier. Best: in the best set. Lead: rival lead, how far the strongest cheaper-or-equal system sits above this one, in paired standard errors (negative when this system is ahead); the simultaneous margin for this board is 3.09.

System	$/task	Unc.	Acc. %	Front.	Set	Fixed	Rob.	Stab.	Best	Lead
HAL Generalist Agent (g	0.021	0	3.0	yes	yes	yes	yes	0.84	yes	
Assistantbench Browser (1)	0.028	0	0.0		yes	yes	yes	0.08	yes	+1.0
Hal Generalist Agent (o (1)	0.029	0	3.0		yes	yes	yes	0.15	yes	+0.0
Assistantbench Browser (2)	0.041	0	0.0		yes	yes	yes	0.00	yes	+1.0
HAL Generalist Agent (d (1)	0.060	0	12.1	yes	yes	yes	yes	0.84	yes	-1.4
Assistantbench Browser (3)	0.062	0	6.1		yes	yes	yes	0.33	yes	+1.0
Hal Generalist Agent (o (2)	0.073	0	0.0		yes	yes	yes	0.00	yes	+2.1
Hal Generalist Agent (o (3)	0.080	0	3.0		yes	yes	yes	0.00	yes	+1.4
Assistantbench Browser (4)	0.084	0	6.1		yes	yes	yes	0.10	yes	+0.8
Assistantbench Browser (5)	0.089	0	6.1		yes	yes	yes	0.10	yes	+0.8
Assistantbench Browser (6)	0.094	0	6.1		yes	yes	yes	0.03	yes	+1.0
Assistantbench Browser (7)	0.101	0	0.0		yes	yes	yes	0.00	yes	+2.1
Hal Generalist Agent (g	0.104	0	9.1		yes	yes	yes	0.09	yes	+0.6
Assistantbench Browser (8)	0.120	0	6.1		yes	yes	yes	0.00	yes	+0.8
Hal Generalist Agent (o (4)	0.127	0	6.1		yes	yes	yes	0.01	yes	+1.0
Assistantbench Browser (9)	0.293	0	0.0		yes	yes	yes	0.00	yes	+2.1
HAL Generalist Agent (c (1)	0.294	0	0.0		yes	yes	yes	0.00	yes	+2.1
Assistantbench Browser (10)	0.295	0	0.0		yes	yes	yes	0.00	yes	+2.1
HAL Generalist Agent (c (2)	0.297	0	3.0		yes	yes	yes	0.00	yes	+1.4
HAL Generalist Agent (c (3)	0.320	0	3.0		yes	yes	yes	0.00	yes	+1.4
HAL Generalist Agent (d (2)	0.370	0	0.0		yes	yes	yes	0.00	yes	+2.1
Assistantbench Browser (11)	0.434	0	3.0		yes	yes	yes	0.02	yes	+1.4
Assistantbench Browser (12)	0.451	0	9.1		yes	yes	yes	0.14	yes	+0.4
Assistantbench Browser (13)	0.489	0	0.0		yes	yes	yes	0.00	yes	+2.1
Assistantbench Browser (14)	0.574	0	3.0		yes	yes	yes	0.00	yes	+1.8
Assistantbench Browser (15)	0.756	0	6.1		yes	yes	yes	0.03	yes	+1.0
Assistantbench Browser (16)	1.107	0	0.0		yes	yes	yes	0.00	yes	+2.1
Assistantbench Browser (17)	2.983	0	3.0		yes	yes	yes	0.00	yes	+1.8
HAL Generalist Agent (c (4)	3.287	0	6.1		yes	yes	yes	0.03	yes	+0.8
HAL Generalist Agent (c (5)	3.412	0	6.1		yes	yes	yes	0.03	yes	+0.8
HAL Generalist Agent (c (6)	4.394	0	15.2	yes	yes	yes	yes	0.42	yes	-0.6
Assistantbench Browser (18)	10.094	0	0.0		yes	yes	yes	0.00	yes	+2.4

Table S2: ColBench backend programming: 22 systems on 1000 tasks, cheapest first. Unc.: tasks with an outcome but no logged model call, priced at the run's mean costed task. Front.: on the observed frontier. Set: in the frontier set. Fixed: in the frontier set with every cost fixed at its mean. Rob.: in the tie-robust set. Stab.: share of task resamples in which the system is on the observed frontier. Best: in the best set. Lead: rival lead, how far the strongest cheaper-or-equal system sits above this one, in paired standard errors (negative when this system is ahead); the simultaneous margin for this board is 3.36.

System	$/task	Unc.	Acc. %	Front.	Set	Fixed	Rob.	Stab.	Best	Lead
colbench_text_deepseekv3	0.002	202	21.0	yes	yes	yes	yes	1.00		
colbench_backend_progra (1)	0.002	0	24.5	yes	yes	yes	yes	1.00		-3.0
colbench_text_gemmini2f	0.006	0	32.9	yes	yes	yes	yes	0.93		-6.0
colbench_backend_progra (2)	0.006	0	33.8	yes	yes	yes	yes	0.76		-0.7
colbench_text_deepseekr1	0.007	377	18.9					0.00		+10.4
colbench_text_o4mini	0.009	0	30.9		yes	yes	yes	0.01		+2.0
colbench_text_gpt41	0.014	0	35.6	yes	yes	yes	yes	0.86		-1.2
colbench_backend_progra (3)	0.015	0	37.2	yes	yes	yes	yes	0.83		-1.0
colbench_example_agent_	0.016	0	38.6	yes	yes	yes	yes	0.81	yes	-0.9
colbench_backend_progra (4)	0.024	0	41.2	yes	yes	yes	yes	0.95	yes	-1.7
colbench_backend_progra (5)	0.026	0	37.1		yes	yes	yes	0.00		+2.7
Hal Generalist Agent (D (1)	0.030	0	34.5					0.00		+4.6
colbench_backend_progra (6)	0.031	0	34.3					0.00		+4.6
colbench_text_sonnet37	0.034	0	37.4		yes	yes	yes	0.00		+2.8
colbench_backend_progra (7)	0.035	0	40.7		yes	yes	yes	0.35	yes	+0.4
colbench_backend_progra (8)	0.041	0	40.0		yes	yes	yes	0.10	yes	+0.9
colbench_backend_progra (9)	0.053	1	22.3					0.00		+12.2
Hal Generalist Agent (G	0.056	0	35.9				yes	0.00		+3.5
Hal Generalist Agent (O (1)	0.063	0	30.4					0.00		+5.0
Hal Generalist Agent (D (2)	0.106	0	29.6					0.00		+5.4
Hal Generalist Agent (O (2)	0.106	0	25.0					0.00		+7.9
Hal Generalist Agent (S	0.202	0	43.7	yes	yes	yes	yes	0.82	yes	-1.2

Table S3: ColBench frontend design: 14 systems on 100 tasks, cheapest first. Unc.: tasks with an outcome but no logged model call, priced at the run's mean costed task. Front.: on the observed frontier. Set: in the frontier set. Fixed: in the frontier set with every cost fixed at its mean. Rob.: in the tie-robust set. Stab.: share of task resamples in which the system is on the observed frontier. Best: in the best set. Lead: rival lead, how far the strongest cheaper-or-equal system sits above this one, in paired standard errors (negative when this system is ahead); the simultaneous margin for this board is 3.14.

System	$/task	Unc.	Acc. %	Front.	Set	Fixed	Rob.	Stab.	Best	Lead
colbench_text_deepseekr1	0.009	2	74.9	yes	yes	yes	yes	0.97		
colbench_text_deepseekv3	0.010	16	75.2	yes	yes	yes	yes	0.66		-0.3
colbench_text_gemini	0.027	0	81.6	yes	yes	yes	yes	1.00	yes	-5.0
Hal Generalist Agent (D (1)	0.060	0	71.3					0.00		+7.7
colbench_text_o4minilow	0.061	0	81.0		yes	yes	yes	0.25	yes	+0.7
Hal Generalist Agent (G	0.065	0	72.2					0.00		+8.2
Hal Generalist Agent (o (1)	0.097	0	72.1					0.00		+7.7
colbench_text_o4minimed	0.124	0	82.9	yes	yes	yes	yes	0.90	yes	-1.3
Hal Generalist Agent (o (2)	0.132	0	67.4					0.00		+15.1
colbench_text_o4minihigh	0.142	0	82.5		yes	yes	yes	0.35	yes	+0.4
Hal Generalist Agent (o (3)	0.144	0	70.7					0.00		+10.7
Hal Generalist Agent (D (2)	0.172	0	57.0					0.00		+16.8
colbench_text_sonnet	0.713	0	82.8		yes	yes	yes	0.38	yes	+0.1
Hal Generalist Agent (S	1.048	0	69.3					0.00		+8.7

Table S4: CORE-Bench hard: 60 systems on 45 tasks, cheapest first. Unc.: tasks with an outcome but no logged model call, priced at the run's mean costed task. Front.: on the observed frontier. Set: in the frontier set. Fixed: in the frontier set with every cost fixed at its mean. Rob.: in the tie-robust set. Stab.: share of task resamples in which the system is on the observed frontier. Best: in the best set. Lead: rival lead, how far the strongest cheaper-or-equal system sits above this one, in paired standard errors (negative when this system is ahead); the simultaneous margin for this board is 3.77.

System	$/task	Unc.	Acc. %	Front.	Set	Fixed	Rob.	Stab.	Best	Lead
HAL Generalist Agent(De (1)	0.017	0	8.9	yes	yes	yes	yes	1.00		
HAL Generalist Agent (1)	0.046	0	8.9		yes	yes	yes	0.42		+0.0
HAL Generalist Agent(De (2)	0.057	0	8.9		yes	yes	yes	0.26		+0.0
HAL Generalist Agent (2)	0.062	0	8.9		yes	yes	yes	0.19		+0.0
Hal Generalist Agent(Ge	0.078	0	4.4		yes	yes	yes	0.01		+1.0
CORE-Agent (1)	0.093	0	11.1	yes	yes	yes	yes	0.26		-0.3

System	$/task	Unc.	Acc. %	Front.	Set	Fixed	Rob.	Stab.	Best	Lead
CORE-Agent (2)	0.095	1	11.1		yes	yes	yes	0.21		+0.0
HAL Generalist Agent(o4 (1)	0.121	0	15.6	yes	yes	yes	yes	0.35		-0.7
CORE-Agent(Deepseek-V3.	0.136	2	20.0	yes	yes	yes	yes	0.59		-0.6
CORE-Agent (3)	0.138	0	11.1		yes	yes	yes	0.07		+1.4
coreagent	0.159	0	20.0		yes	yes	yes	0.41		+0.0
HAL Generalist Agent (3)	0.190	0	15.6		yes	yes	yes	0.07		+0.7
HAL Generalist Agent (4)	0.196	0	6.7		yes	yes	yes	0.00		+2.2
HAL Generalist Agent (5)	0.209	0	15.6		yes	yes	yes	0.04		+0.7
HAL Generalist Agent (6)	0.210	1	8.9		yes	yes	yes	0.00		+1.9
CORE-Agent (4)	0.216	5	15.6		yes	yes	yes	0.02		+1.0
CORE-Agent (5)	0.267	0	17.8		yes	yes	yes	0.07		+0.4
CORE-Agent (6)	0.281	0	17.8		yes	yes	yes	0.06		+0.4
HAL Generalist Agent(gp (1)	0.296	0	22.2	yes	yes	yes	yes	0.19		-0.3
Hal Generalist Agent(De	0.308	0	2.2		yes	yes	yes	0.00		+3.1
HAL Generalist Agent(gp (2)	0.320	1	11.1		yes	yes	yes	0.00		+1.7
HAL Generalist Agent (7)	0.325	2	13.3		yes	yes	yes	0.00		+1.4
Core Agent	0.338	0	40.0	yes	yes	yes	yes	0.83		-1.9
HAL Generalist Agent(o4 (2)	0.365	0	35.6		yes	yes	yes	0.47		+0.4
HAL Generalist Agent (8)	0.395	1	35.6		yes	yes	yes	0.30		+0.6
HAL Generalist Agent (9)	0.447	11	4.4				yes	0.00		+4.5
HAL Generalist Agent(o3	0.470	1	22.2		yes	yes	yes	0.01		+1.8
HAL Generalist Agent (10)	0.504	0	24.4		yes	yes	yes	0.00		+1.9
HAL Generalist Agent (11)	0.523	0	6.7				yes	0.00		+4.2
CORE-Agent (7)	0.527	0	33.3		yes	yes	yes	0.07		+0.7
CORE-Agent (8)	0.546	0	26.7		yes	yes	yes	0.01		+1.4
CORE-Agent (9)	0.692	0	24.4		yes	yes	yes	0.00		+1.6
Core-Agent(Deepseek-R1)	0.788	1	4.4				yes	0.00		+4.9
HAL Generalist Agent (12)	0.815	0	8.9		yes	yes	yes	0.00		+3.7
CORE-Agent(Claude-Haiku	0.976	0	11.1		yes	yes	yes	0.00		+3.3
CORE-Agent(Claude-Sonne (1)	1.140	1	28.9		yes	yes	yes	0.02		+1.0
HAL Generalist Agent (13)	1.146	0	26.7		yes	yes	yes	0.01		+1.4
HAL Generalist Agent (14)	1.236	1	31.1		yes	yes	yes	0.03		+1.0
HAL Generalist Agent(so (1)	1.258	0	31.1		yes	yes	yes	0.02		+1.1
HAL Generalist Agent (15)	1.405	0	22.2		yes	yes	yes	0.00		+1.8
CORE-Agent (10)	1.450	1	8.9				yes	0.00		+4.0
HAL Generalist Agent(so (2)	1.469	0	37.8		yes	yes	yes	0.17		+0.2
CORE-Agent (11)	1.609	0	24.4		yes	yes	yes	0.00		+1.9
CORE-Agent (12)	1.622	0	35.6		yes	yes	yes	0.07		+0.5
CORE-Agent (13)	1.671	0	33.3		yes	yes	yes	0.04		+0.7
HAL-Generalist-Agent(Cl (1)	1.892	0	33.3		yes	yes	yes	0.04		+0.8
HAL-Generalist-Agent(Cl (2)	1.950	0	28.9		yes	yes	yes	0.00		+1.4
CORE-Agent (14)	2.026	0	22.2		yes	yes	yes	0.00		+2.0
CORE-Agent(Claude-Sonne (2)	2.099	1	44.4	yes	yes	yes	yes	0.43		-0.5
CORE-Agent(Claude- Sonn	2.208	1	37.8		yes	yes	yes	0.11		+0.8
CORE-Agent(Claude-Sonne (3)	2.232	0	33.3		yes	yes	yes	0.00		+1.5
Hal Generalist Agent Op (1)	2.554	1	31.1		yes	yes	yes	0.02		+1.3
Hal Generalist Agent Op (2)	2.831	0	33.3		yes	yes	yes	0.02		+1.2
Core Agent Opus 4.5 high	3.390	0	42.2		yes	yes	yes	0.17		+0.3
Core Agent Opus 4.5	3.755	0	42.2		yes	yes	yes	0.12		+0.3
Coreagent(claude-opus-4	6.082	0	75.6	yes	yes	yes	yes	1.00	yes	-3.7
HAL Generalist Agent (16)	8.147	1	33.3				yes	0.00		+5.7
HAL Generalist Agent (17)	8.525	1	35.6				yes	0.00		+5.0
CORE-Agent (15)	9.164	0	51.1		yes	yes	yes	0.01		+3.4
CORE-Agent (16)	11.331	0	42.2				yes	0.00		+4.0

Table S5: GAIA: 36 systems on 165 tasks, cheapest first. Unc.: tasks with an outcome but no logged model call, priced at the run's mean costed task. Front.: on the observed frontier. Set: in the frontier set. Fixed: in the frontier set with every cost fixed at its mean. Rob.: in the tie-robust set. Stab.: share of task resamples in which the system is on the observed frontier. Best: in the best set. Lead: rival lead, how far the strongest cheaper-or-equal system sits above this one, in paired standard errors (negative when this system is ahead); the simultaneous margin for this board is 3.65.

System	$/task	Unc.	Acc. %	Front.	Set	Fixed	Rob.	Stab.	Best	Lead
HAL Generalist Agent (d (1)	0.006	0	22.4	yes	yes	yes	yes	1.00		
HAL Generalist Agent (g (1)	0.024	0	32.7	yes	yes	yes	yes	0.98		-2.7
HAL Generalist Agent (o (1)	0.032	0	33.9	yes	yes	yes	yes	0.59		-0.3
HF Open Deep Research ((1)	0.057	0	19.4				yes	0.00		+3.7
HAL Generalist Agent (o (2)	0.086	0	46.7	yes	yes	yes	yes	0.99		-3.1
HAL Generalist Agent (D	0.089	0	36.4		yes	yes	yes	0.30		+2.2
HAL Generalist Agent (g (2)	0.121	0	49.7	yes	yes	yes	yes	0.63		-0.6
HAL Generalist Agent (o (3)	0.145	0	54.5	yes	yes	yes	yes	0.71		-1.0
HF Open Deep Research ((2)	0.154	1	18.2					0.00		+7.8
HAL Generalist Agent (o (4)	0.175	0	58.2	yes	yes	yes	yes	0.75		-1.0
HF Open Deep Research ((3)	0.190	1	50.3		yes	yes	yes	0.06		+1.6
HF Open Deep Research ((4)	0.195	1	47.9		yes	yes	yes	0.01		+2.2

System	$/task	Unc.	Acc. %	Front.	Set	Fixed	Rob.	Stab.	Best	Lead
HAL Generalist Agent (g (3)	0.219	0	59.4	yes	yes	yes	yes	0.55		-0.3
HAL Generalist Agent (d (2)	0.222	0	30.3				yes	0.00		+6.6
HF Open Deep Research ((5)	0.232	0	28.5					0.00		+6.6
HF Open Deep Research ((6)	0.283	0	32.7					0.00		+5.6
HF Open Deep Research ((7)	0.434	0	24.8					0.00		+7.4
HF Open Deep Research ((8)	0.482	0	55.8		yes	yes	yes	0.11		+0.8
HF Open Deep Research ((9)	0.689	0	35.8					0.00		+4.9
HAL Generalist Agent (c (1)	0.742	0	64.2	yes	yes	yes	yes	0.72	yes	-1.0
HAL Generalist Agent (c (2)	0.791	0	56.4		yes	yes	yes	0.04		+1.8
HAL Generalist Agent (C	0.843	10	56.4		yes	yes	yes	0.05		+1.6
HAL Generalist Agent (c (3)	1.023	0	74.5	yes	yes	yes	yes	0.85	yes	-2.4
HF Open Deep Research ((10)	1.024	0	62.4		yes	yes	yes	0.11		+3.2
HAL Generalist Agent (c (4)	1.089	0	70.9		yes	yes	yes	0.25	yes	+1.2
HAL Generalist Agent (c (5)	1.134	0	74.5		yes	yes	yes	0.56	yes	+0.0
HF Open Deep Research ((11)	2.678	10	37.0					0.00		+8.4
HAL Generalist Agent (c (6)	2.700	64	30.3					0.00		+10.4
HF Open Deep Research ((12)	2.724	0	30.9					0.00		+10.8
HF Open Deep Research ((13)	3.228	0	30.9					0.00		+9.9
HAL Generalist Agent (c (7)	3.407	0	68.5		yes	yes	yes	0.00	yes	+1.9
HAL Generalist Agent (c (8)	3.889	0	64.2		yes	yes	yes	0.00	yes	+2.9
HAL Generalist Agent (c (9)	4.059	1	64.8		yes	yes	yes	0.00	yes	+2.6
HF Open Deep Research ((14)	7.920	0	28.5					0.00		+11.3
HF Open Deep Research ((15)	8.931	0	25.5					0.00		+12.3
HF Open Deep Research ((16)	10.219	0	57.6					0.00		+4.1

Table S6: Online-Mind2Web†: 32 systems on 300 tasks, cheapest first. Unc.: tasks with an outcome but no logged model call, priced at the run's mean costed task. Front.: on the observed frontier. Set: in the frontier set. Fixed: in the frontier set with every cost fixed at its mean. Rob.: in the tie-robust set. Stab.: share of task resamples in which the system is on the observed frontier. Best: in the best set. Lead: rival lead, how far the strongest cheaper-or-equal system sits above this one, in paired standard errors (negative when this system is ahead); the simultaneous margin for this board is 3.56.

System	$/task	Unc.	Acc. %	Front.	Set	Fixed	Rob.	Stab.	Best	Lead
SeeAct(gemini-2.0-flash)	0.017	0	26.7	yes	yes	yes	yes	1.00		
Browser-Use_test(gemini (1)	0.060	0	29.0	yes	yes	yes	yes	0.78		-0.9
SeeAct(claude-haiku-4-5 (1)	0.223	0	32.7	yes	yes	yes	yes	0.87		-1.2
SeeAct(gpt-5-2025-08-07 (1)	0.263	0	30.3		yes	yes	yes	0.17		+0.8
SeeAct(claude-haiku-4-5 (2)	0.297	0	36.3	yes	yes	yes	yes	0.92	yes	-1.5
SeeAct(gemini-2.5-pro-p	0.512	0	39.7	yes	yes	yes	yes	0.85	yes	-1.1
SeeAct(o4-mini-2025-04- (1)	0.541	0	31.7		yes	yes	yes	0.00		+2.7
SeeAct(gpt-5-2025-08-07 (2)	0.570	0	42.3	yes	yes	yes	yes	0.82	yes	-1.0
Browser-Use(o4-mini-202 (1)	0.671	0	18.3					0.00		+7.6
SeeAct(claude-sonnet-4- (1)	0.683	0	35.0		yes	yes	yes	0.00	yes	+2.4
Browser-Use_test(DeepSe (1)	0.716	0	32.3		yes	yes	yes	0.00		+3.3
SeeAct(o4-mini-2025-04- (2)	0.763	0	32.0					0.00		+3.8
Browser-Use(gpt-4.1-202	0.789	0	36.3		yes	yes	yes	0.03	yes	+1.8
SeeAct(claude-sonnet-4- (2)	0.821	0	36.7		yes	yes	yes	0.01	yes	+2.0
SeeAct(o3-2025-04-16)	0.862	0	39.0		yes	yes	yes	0.08	yes	+1.2
SeeAct(gpt-4.1-2025-04-	0.904	0	30.3					0.00		+3.9
Browser-Use_test(DeepSe (2)	0.936	0	25.3					0.00		+5.0
SeeAct(claude-3-7-sonne (1)	0.973	0	28.3					0.00		+5.1
SeeAct(claude-sonnet-4- (3)	0.977	0	38.0		yes	yes	yes	0.03	yes	+1.4
Browser-Use(o4-mini-202 (2)	0.993	0	20.0					0.00		+7.0
SeeAct(claude-sonnet-4- (4)	1.088	0	36.7		yes	yes	yes	0.01	yes	+1.9
Browser-Use_test(gemini (2)	1.105	0	41.0		yes	yes	yes	0.26	yes	+0.4
SeeAct(gpt-5-2025-08-07 (3)	1.143	0	41.7		yes	yes	yes	0.22	yes	+0.3
Browser-Use(gpt-5-2025- (1)	1.157	0	22.0					0.00		+6.9
SeeAct(claude-3-7-sonne (2)	1.225	0	30.3					0.00		+4.2
Browser-Use(o3-2025-04-	1.239	0	29.0					0.00		+4.1
Browser-Use(gpt-5-2025- (2)	2.454	0	32.0		yes	yes	yes	0.00		+3.2
Browser-Use_test(claude (1)	3.088	0	38.3		yes	yes	yes	0.03	yes	+1.2
Browser-Use(gpt-5-2025- (3)	3.565	0	35.7		yes	yes	yes	0.00	yes	+1.9
Browser-Use_test(claude (2)	3.840	0	39.3		yes	yes	yes	0.06	yes	+0.9
Browser-Use_test(claude (3)	5.258	0	40.0		yes	yes	yes	0.07	yes	+0.7
Browser-Use_test(claude (4)	5.366	0	39.3		yes	yes	yes	0.03	yes	+0.9

Table S7: SciCode: 37 systems on 65 tasks, cheapest first. Unc.: tasks with an outcome but no logged model call, priced at the run's mean costed task. Front.: on the observed frontier. Set: in the frontier set. Fixed: in the frontier set with every cost fixed at its mean. Rob.: in the tie-robust set. Stab.: share of task resamples in which the system is on the observed frontier. Best: in the best set. Lead: rival lead, how far the strongest cheaper-or-equal system sits above this one, in paired standard errors (negative when this system is ahead); the simultaneous margin for this board is 3.56.

System	$/task	Unc.	Acc. %	Front.	Set	Fixed	Rob.	Stab.	Best	Lead
Scicode Zero Shot Agent (1)	0.002	0	1.5	yes	yes	yes	yes	1.00	yes	
Scicode Zero Shot Agent (2)	0.012	0	3.1	yes	yes	yes	yes	0.63	yes	-1.0
Scicode Zero Shot Agent (3)	0.026	0	9.2	yes	yes	yes	yes	0.98	yes	-2.0
Scicode Zero Shot Agent (4)	0.039	0	6.2		yes	yes	yes	0.19	yes	+0.7
Scicode Tool Calling Ag (1)	0.041	0	1.5		yes	yes	yes	0.00	yes	+2.3
Scicode Zero Shot Agent (5)	0.042	0	6.2		yes	yes	yes	0.07	yes	+1.0
Scicode Zero Shot Agent (6)	0.077	0	3.1		yes	yes	yes	0.00	yes	+1.7
Scicode Zero Shot Agent (7)	0.078	0	0.0		yes	yes	yes	0.00	yes	+2.6
Scicode Zero Shot Agent (8)	0.083	1	6.2		yes	yes	yes	0.04	yes	+1.0
Scicode Zero Shot Agent (9)	0.089	0	4.6		yes	yes	yes	0.01	yes	+1.4
Scicode Zero Shot Agent (10)	0.143	1	0.0		yes	yes	yes	0.00	yes	+2.6
Scicode Tool Calling Ag (2)	0.320	0	4.6		yes	yes	yes	0.00	yes	+1.8
Hal Generalist Agent (o (1)	0.355	0	6.2		yes	yes	yes	0.06	yes	+0.8
Hal Generalist Agent (g	0.359	0	1.5		yes	yes	yes	0.00	yes	+1.9
Hal Generalist Agent (o (2)	0.382	0	3.1		yes	yes	yes	0.00	yes	+1.7
Scicode Tool Calling Ag (3)	0.394	0	1.5		yes	yes	yes	0.00	yes	+2.3
Scicode Tool Calling Ag (4)	0.399	0	3.1		yes	yes	yes	0.00	yes	+2.0
Scicode Tool Calling Ag (5)	0.404	0	0.0		yes	yes	yes	0.00	yes	+2.6
Scicode Tool Calling Ag (6)	0.494	0	4.6		yes	yes	yes	0.00	yes	+1.8
Hal Generalist Agent (o (3)	0.576	0	1.5		yes	yes	yes	0.00	yes	+1.9
Hal Generalist Agent (o (4)	0.623	1	1.5		yes	yes	yes	0.00	yes	+2.3
Scicode Tool Calling Ag (7)	0.722	0	9.2		yes	yes	yes	0.26	yes	+0.0
HAL Generalist Agent (c (1)	0.990	4	3.1		yes	yes	yes	0.00	yes	+1.7
Hal Generalist Agent (o (5)	1.029	0	6.2		yes	yes	yes	0.01	yes	+1.0
Scicode Tool Calling Ag (8)	1.442	0	6.2		yes	yes	yes	0.01	yes	+1.0
Scicode Tool Calling Ag (9)	1.499	0	3.1		yes	yes	yes	0.00	yes	+2.0
Scicode Tool Calling Ag (10)	1.599	0	4.6		yes	yes	yes	0.01	yes	+1.4
Scicode Tool Calling Ag (11)	1.604	0	3.1		yes	yes	yes	0.00	yes	+2.0
HAL Generalist Agent (d (1)	1.714	1	0.0		yes	yes	yes	0.00	yes	+2.6
Scicode Tool Calling Ag (12)	1.807	0	0.0		yes	yes	yes	0.00	yes	+2.6
Scicode Tool Calling Ag (13)	1.818	0	1.5		yes	yes	yes	0.00	yes	+2.3
Scicode Tool Calling Ag (14)	2.142	0	0.0		yes	yes	yes	0.00	yes	+2.6
HAL Generalist Agent (c (2)	2.895	0	3.1		yes	yes	yes	0.01	yes	+1.4
HAL Generalist Agent (d (2)	3.744	0	0.0		yes	yes	yes	0.00	yes	+2.6
SciCode Tool Calling Ag	7.811	0	7.7		yes	yes	yes	0.06	yes	+0.6
Scicode Tool Calling Ag (15)	9.128	0	6.2		yes	yes	yes	0.00	yes	+1.4
Scicode Tool Calling Ag (16)	9.617	0	7.7		yes	yes	yes	0.04	yes	+0.6

Table S8: ScienceAgentBench: 25 systems on 102 tasks, cheapest first. Unc.: tasks with an outcome but no logged model call, priced at the run's mean costed task. Front.: on the observed frontier. Set: in the frontier set. Fixed: in the frontier set with every cost fixed at its mean. Rob.: in the tie-robust set. Stab.: share of task resamples in which the system is on the observed frontier. Best: in the best set. Lead: rival lead, how far the strongest cheaper-or-equal system sits above this one, in paired standard errors (negative when this system is ahead); the simultaneous margin for this board is 3.46.

System	$/task	Unc.	Acc. %	Front.	Set	Fixed	Rob.	Stab.	Best	Lead
SAB Self-Debug gemini-2 (1)	0.002	0	12.7	yes	yes	yes	yes	1.00		
SAB Self-Debug DS-V3	0.010	0	15.7	yes	yes	yes	yes	0.80		-1.0
SAB Self-Debug o4-mini (1)	0.019	0	27.5	yes	yes	yes	yes	0.99	yes	-2.4
SAB Self-Debug Claude-H (1)	0.026	0	18.6		yes	yes	yes	0.04	yes	+1.7
SAB Self-Debug Claude-H (2)	0.033	0	23.5		yes	yes	yes	0.15	yes	+0.8
sab_selfdebug_gpt_41	0.034	0	24.5		yes	yes	yes	0.20	yes	+0.6
SAB Self-Debug o4-mini (2)	0.055	0	27.5		yes	yes	yes	0.27	yes	+0.0
SAB Self-Debug o3 medium	0.057	0	33.3	yes	yes	yes	yes	0.77	yes	-1.2
SAB Self-Debug Claude-3 (1)	0.064	0	25.5		yes	yes	yes	0.04	yes	+1.6
SAB Self-Debug Claude-3 (2)	0.070	0	22.5		yes	yes	yes	0.01	yes	+2.2
SAB Self-Debug Claude-S (1)	0.072	0	29.4		yes	yes	yes	0.11	yes	+0.9
SAB Self-Debug Claude-S (2)	0.073	0	30.4		yes	yes	yes	0.18	yes	+0.6
SAB Self-Debug GPT-5-Me	0.089	0	30.4		yes	yes	yes	0.08	yes	+0.8
sab_selfdebug_dsr1	0.089	0	23.5		yes	yes	yes	0.00	yes	+2.0
SAB Self-Debug gemini-2 (2)	0.110	0	18.6		yes	yes	yes	0.00	yes	+2.8
SAB Self-Debug Claude-3 (3)	0.115	0	30.4		yes	yes	yes	0.09	yes	+0.7
HAL Generalist Agent (o (1)	0.138	0	9.8					0.00		+4.5
HAL Generalist Agent (g	0.232	0	6.9					0.00		+5.3
HAL Generalist Agent (d	0.273	0	1.0					0.00		+6.7
SAB Self-Debug Claude-O (1)	0.327	0	27.5		yes	yes	yes	0.03	yes	+1.2

System	$/task	Unc.	Acc. %	Front.	Set	Fixed	Rob.	Stab.	Best	Lead
SAB Self-Debug Claude-O (2)	0.331	0	26.5		yes	yes	yes	0.02	yes	+1.5
HAL Generalist Agent (o (2)	0.331	0	21.6		yes	yes	yes	0.00	yes	+2.4
HAL Generalist Agent (o (3)	0.336	0	19.6		yes	yes	yes	0.00		+3.1
HAL Generalist Agent (c (1)	0.404	0	10.8					0.00		+5.0
HAL Generalist Agent (c (2)	0.473	0	17.6		yes	yes	yes	0.00		+2.9

Table S9: SWE-bench Verified mini: 39 systems on 50 tasks, cheapest first. Unc.: tasks with an outcome but no logged model call, priced at the run's mean costed task. Front.: on the observed frontier. Set: in the frontier set. Fixed: in the frontier set with every cost fixed at its mean. Rob.: in the tie-robust set. Stab.: share of task resamples in which the system is on the observed frontier. Best: in the best set. Lead: rival lead, how far the strongest cheaper-or-equal system sits above this one, in paired standard errors (negative when this system is ahead); the simultaneous margin for this board is 3.66.

System	$/task	Unc.	Acc. %	Front.	Set	Fixed	Rob.	Stab.	Best	Lead
hal_generalist_agent_ge	0.073	0	2.0	yes	yes	yes	yes	0.63		
hal_generalist_agent_o3 (1)	0.075	0	0.0		yes	yes	yes	0.20		+1.0
hal_generalist_agent_o3 (2)	0.077	0	0.0		yes	yes	yes	0.40		+1.0
My Agent(gemini/gemini-	0.094	0	22.0	yes	yes	yes	yes	1.00		-3.1
HAL Generalist Agent (o (1)	0.229	0	2.0		yes	yes	yes	0.00		+3.5
hal_generalist_agent_de (1)	0.302	0	10.0		yes	yes	yes	0.02		+1.9
HAL Generalist Agent (g (1)	0.324	0	2.0		yes	yes	yes	0.00		+3.5
HAL Generalist Agent (o (2)	0.398	0	6.0		yes	yes	yes	0.01		+2.7
HAL Generalist (gpt-5-2	0.451	0	14.0		yes	yes	yes	0.09		+1.2
SWE-Agent(claude-3-7-so (1)	0.667	0	54.0	yes	yes	yes	yes	0.72	yes	-4.4
SWE-Agent(claude-3-7-so (2)	0.681	0	52.0		yes	yes	yes	0.53		+0.4
My Agent(claude-3-7-son	0.794	0	50.0		yes	yes	yes	0.11		+0.7
SWE-Agent(gpt-5-2025-08	1.020	0	64.0	yes	yes	yes	yes	0.89	yes	-1.4
My Agent(o3-mini)	1.212	1	46.0		yes	yes	yes	0.02		+2.4
SWE-Agent(gpt-4.1)	1.265	0	46.0		yes	yes	yes	0.00		+2.9
HAL Generalist (claude- (1)	1.306	0	44.0		yes	yes	yes	0.00		+2.6
hal_generalist_agent_de (2)	1.467	0	6.0				yes	0.00		+7.6
HAL Generalist Agent (g (2)	1.496	0	2.0					0.00		+8.9
SWE-Agent(claude-sonnet (1)	1.509	0	68.0	yes	yes	yes	yes	0.59	yes	-0.7
HAL Generalist (claude- (2)	1.536	0	38.0				yes	0.00		+4.6
SWE-Agent(claude-sonnet (2)	1.573	0	72.0	yes	yes	yes	yes	0.75	yes	-1.0
HAL Generalist (claude- (3)	1.744	0	24.0				yes	0.00		+6.7
HAL Generalist (o3-2025	1.884	0	0.0				yes	0.00		+11.2
HAL Generalist (claude- (4)	1.999	2	40.0				yes	0.00		+4.4
My Agent(claude-opus-4-	2.119	0	60.0		yes	yes	yes	0.01	yes	+1.8
HAL Generalist Agent (c	2.140	1	26.0					0.00		+6.5
SWE-Agent(o4-mini-2025- (1)	2.318	1	52.0		yes	yes	yes	0.00		+3.1
SWE-Agent(o4-mini-2025- (2)	2.445	2	56.0		yes	yes	yes	0.00	yes	+2.4
HAL Generalist (claude- (5)	2.616	1	34.0					0.00		+5.1
SWE-Agent(claude-opus-4 (1)	2.623	0	66.0		yes	yes	yes	0.10	yes	+0.9
SWE-Agent(claude-opus-4 (2)	2.811	0	54.0		yes	yes	yes	0.00	yes	+2.3
SWE-Agent(claude-opus-4 (3)	3.403	0	68.0		yes	yes	yes	0.13	yes	+0.6
My Agent(gpt-4o)	3.507	0	28.0					0.00		+5.4
HAL Generalist (claude- (6)	4.171	0	24.0					0.00		+6.2
My Agent(o1)	4.478	0	40.0				yes	0.00		+4.4
HAL Generalist (claude- (7)	7.648	0	34.0					0.00		+5.5
HAL Generalist (claude- (8)	7.999	0	56.0		yes	yes	yes	0.00	yes	+2.7
HAL Generalist (claude- (9)	8.068	0	30.0					0.00		+5.5
HAL Generalist (claude- (10)	9.553	0	52.0		yes	yes	yes	0.00	yes	+2.9

Table S10: tau-bench airline: 47 systems on 50 tasks, cheapest first. Unc.: tasks with an outcome but no logged model call, priced at the run's mean costed task. Front.: on the observed frontier. Set: in the frontier set. Fixed: in the frontier set with every cost fixed at its mean. Rob.: in the tie-robust set. Stab.: share of task resamples in which the system is on the observed frontier. Best: in the best set. Lead: rival lead, how far the strongest cheaper-or-equal system sits above this one, in paired standard errors (negative when this system is ahead); the simultaneous margin for this board is 3.67.

System	$/task	Unc.	Acc. %	Front.	Set	Fixed	Rob.	Stab.	Best	Lead
HAL Generalist Agent (g (1)	0.020	0	22.0	yes	yes	yes	yes	1.00		
Taubench ToolCalling (g (1)	0.028	0	28.0	yes	yes	yes	yes	0.71		-0.7
TAU-bench FewShot (gemi	0.044	0	44.0	yes	yes	yes	yes	0.86	yes	-2.1
Taubench ToolCalling (o (1)	0.045	0	36.0		yes	yes	yes	0.40		+0.9
Taubench ToolCalling (g (2)	0.077	0	36.0		yes	yes	yes	0.09		+0.9
Taubench ToolCalling (o (2)	0.084	0	56.0	yes	yes	yes	yes	0.78	yes	-1.3
Taubench ToolCalling (o (3)	0.090	0	54.0		yes	yes	yes	0.45	yes	+0.3

System	$/task	Unc.	Acc. %	Front.	Set	Fixed	Rob.	Stab.	Best	Lead
HAL Generalist Agent (d (1)	0.107	0	18.0				yes	0.00		+5.1
HAL Generalist Agent (g (2)	0.115	0	16.0				yes	0.00		+5.3
TAU-bench FewShot (o3-m (1)	0.128	0	40.0		yes	yes	yes	0.00		+2.4
Taubench ToolCalling (d (1)	0.130	1	44.0		yes	yes	yes	0.02		+1.4
TAU-bench FewShot (o3-m (2)	0.140	0	48.0		yes	yes	yes	0.05	yes	+1.1
TAU-bench FewShot (o4-m (1)	0.145	0	60.0	yes	yes	yes	yes	0.52	yes	-0.5
TAU-bench FewShot (o4-m (2)	0.149	0	48.0		yes	yes	yes	0.03	yes	+1.5
HAL Generalist Agent (o (1)	0.173	0	22.0					0.00		+4.2
HAL Generalist Agent (o (2)	0.174	0	18.0					0.00		+4.9
TAU-bench FewShot (o3-2 (1)	0.178	0	44.0		yes	yes	yes	0.00	yes	+1.9
Taubench ToolCalling (g (3)	0.182	0	48.0		yes	yes	yes	0.02	yes	+1.4
TAU-bench FewShot (gpt- (1)	0.213	0	56.0		yes	yes	yes	0.17	yes	+0.5
TAU-bench FewShot (o3-2 (2)	0.242	0	48.0		yes	yes	yes	0.00	yes	+1.8
taubench_fewshot_o3mini	0.243	0	40.0		yes	yes	yes	0.00		+2.6
TauBench Few-Shot Minim	0.248	0	50.0		yes	yes	yes	0.03	yes	+1.2
TauBench Few-Shot High	0.259	0	52.0		yes	yes	yes	0.02	yes	+1.2
HAL Generalist Agent (o (3)	0.263	0	20.0					0.00		+5.3
Taubench ToolCalling (d (2)	0.283	0	36.0		yes	yes	yes	0.00		+2.9
HAL Generalist Agent (d (2)	0.302	0	10.0					0.00		+6.5
TAU-bench FewShot (deep (1)	0.306	0	34.0				yes	0.00		+3.8
TAU-bench FewShot (deep (2)	0.317	0	36.0		yes	yes	yes	0.00		+3.1
Taubench ToolCalling (c (1)	0.324	0	44.0		yes	yes	yes	0.00	yes	+1.8
Taubench ToolCalling (c (2)	0.338	0	52.0		yes	yes	yes	0.04	yes	+0.9
HAL Generalist High Rea (1)	0.447	0	30.0		yes	yes	yes	0.00		+3.7
TAU-bench FewShot (gpt- (2)	0.465	0	40.0		yes	yes	yes	0.00		+2.2
HAL Generalist Minimal	0.517	0	38.0		yes	yes	yes	0.00		+2.7
HAL Generalist Agent (c (1)	0.677	0	44.0		yes	yes	yes	0.01	yes	+1.7
TAU-bench FewShot (clau (1)	0.712	0	34.0		yes	yes	yes	0.00		+3.1
TAU-bench FewShot (clau (2)	0.726	0	60.0		yes	yes	yes	0.27	yes	+0.0
HAL Generalist Agent (c (2)	0.824	0	56.0		yes	yes	yes	0.10	yes	+0.5
Taubench ToolCalling (c (3)	1.415	0	50.0		yes	yes	yes	0.01	yes	+1.4
Taubench ToolCalling (c (4)	1.518	0	52.0		yes	yes	yes	0.01	yes	+1.2
HAL Generalist High Rea (2)	2.793	0	32.0		yes	yes	yes	0.00		+3.3
TauBench FewShot No Rea	2.942	0	54.0		yes	yes	yes	0.01	yes	+0.9
TauBench FewShot High R	2.986	0	62.0	yes	yes	yes	yes	0.22	yes	-0.3
HAL Generalist (claude-	2.987	0	44.0		yes	yes	yes	0.00	yes	+2.1
HAL Generalist High Rea (3)	2.992	0	44.0		yes	yes	yes	0.00	yes	+2.1
TauBench Few Shot High	3.138	0	66.0	yes	yes	yes	yes	0.44	yes	-0.5
HAL Generalist No Reaso	3.593	0	54.0		yes	yes	yes	0.01	yes	+1.5
TauBench Few Shot (clau	3.633	0	56.0		yes	yes	yes	0.01	yes	+1.2

Table S11: USACO: 12 systems on 307 tasks, cheapest first. Unc.: tasks with an outcome but no logged model call, priced at the run's mean costed task. Front.: on the observed frontier. Set: in the frontier set. Fixed: in the frontier set with every cost fixed at its mean. Rob.: in the tie-robust set. Stab.: share of task resamples in which the system is on the observed frontier. Best: in the best set. Lead: rival lead, how far the strongest cheaper-or-equal system sits above this one, in paired standard errors (negative when this system is ahead); the simultaneous margin for this board is 3.04.

System	$/task	Unc.	Acc. %	Front.	Set	Fixed	Rob.	Stab.	Best	Lead
usaco_episodic (1)	0.020	0	39.1	yes	yes	yes	yes	1.00		
USACO Episodic + Semant (1)	0.034	0	30.9					0.00		+3.3
usaco_episodic (2)	0.037	0	35.2		yes	yes	yes	0.05		+1.6
usaco_episodic (3)	0.046	0	45.0	yes	yes	yes	yes	1.00		-2.7
USACO Episodic + Semant (2)	0.076	18	58.0	yes	yes	yes	yes	1.00		-5.3
USACO Episodic + Semant (3)	0.098	14	46.3					0.00		+4.4
usaco_episodic (4)	0.106	16	49.8					0.00		+3.6
USACO Episodic + Semant (4)	0.113	24	69.7	yes	yes	yes	yes	1.00	yes	-4.7
USACO Episodic + Semant (5)	0.126	0	29.3					0.00		+14.0
usaco_episodic (5)	0.154	47	38.1					0.00		+11.4
USACO Episodic + Semant (6)	0.436	0	51.5					0.00		+7.2
USACO Episodic + Semant (7)	0.450	0	48.2					0.00		+7.9

S2. Certification of frontier entrants

The steps behind the certification table of the main text: the pairwise record claim and, in the last two columns, the composite Pareto certificate.

Table S12: AssistantBench: frontier entrants in arrival order. Holder: the most accurate cheaper-or-equal earlier system. p raw: selective record test; g: submissions on the entrant's date; $p = \min(1, g \cdot p_{\text{raw}})$. Levels under alpha-spending and the online fallback, with the certification decision; the last two columns are the composite-null (intersection-union) p-value after the per-date factor and its decision under alpha-spending.

t	Date	Entrant	Holder	Margin	p raw	g	p	Spending	Cert.	Fallback	Cert.	p_{IU}	Cert.	p_{ex}	Cert.
1	2025-08-11	HAL Generalist Ag	HAL Generalist	+3.0	1.000	1	1.000	0.0304		0.0304		0.633		1.000	

Table S13: ColBench backend programming: frontier entrants in arrival order. Holder: the most accurate cheaper-or-equal earlier system. p raw: selective record test; g: submissions on the entrant's date; $p = \min(1, g \cdot p_{\text{raw}})$. Levels under alpha-spending and the online fallback, with the certification decision; the last two columns are the composite-null (intersection-union) p-value after the per-date factor and its decision under alpha-spending.

t	Date	Entrant	Holder	Margin	p raw	g	p	Spending	Cert.	Fallback	Cert.	p_{IU}	Cert.	p_{ex}	Cert.
1	2025-04-20	colbench__text__gpt	colbench__text__g	+2.7	0.088	1	0.088	0.0304		0.0304		0.080		0.088	
2	2025-05-23	Hal Generalist Ag	colbench__text__s	+6.3	0.004	1	0.004	0.0076	yes	0.0076	yes	0.004	yes	0.004	yes
3	2025-09-22	colbench__backend__	colbench__text__g	+5.6	0.000	9	0.002	0.0034	yes	0.0110	yes	0.002	yes	0.002	yes
3	2025-09-22	colbench__example__	colbench__text__g	+3.0	0.052	9	0.469	0.0034		0.0110		0.398		0.421	
3	2025-09-22	colbench__backend__	colbench__text__g	+1.6	0.367	9	1.000	0.0034		0.0110		1.000		1.000	
3	2025-09-22	colbench__backend__	colbench__text__g	+0.9	0.548	9	1.000	0.0034		0.0110		1.000		1.000	
4	2025-09-23	colbench__backend__	colbench__text__d	+3.5	0.004	1	0.004	0.0019		0.0019		0.003		0.004	

Table S14: ColBench frontend design: frontier entrants in arrival order. Holder: the most accurate cheaper-or-equal earlier system. p raw: selective record test; g: submissions on the entrant's date; $p = \min(1, g \cdot p_{\text{raw}})$. Levels under alpha-spending and the online fallback, with the certification decision; the last two columns are the composite-null (intersection-union) p-value after the per-date factor and its decision under alpha-spending.

t	Date	Entrant	Holder	Margin	p raw	g	p	Spending	Cert.	Fallback	Cert.	p_{IU}	Cert.	p_{ex}	Cert.
1	2025-04-23	colbench__text__dee	colbench__text__d	+0.3	0.755	1	0.755	0.0304		0.0304		0.823		0.757	

Table S15: CORE-Bench hard: frontier entrants in arrival order. Holder: the most accurate cheaper-or-equal earlier system. p raw: selective record test; g: submissions on the entrant's date; $p = \min(1, g \cdot p_{\text{raw}})$. Levels under alpha-spending and the online fallback, with the certification decision; the last two columns are the composite-null (intersection-union) p-value after the per-date factor and its decision under alpha-spending.

t	Date	Entrant	Holder	Margin	p raw	g	p	Spending	Cert.	Fallback	Cert.	p_{IU}	Cert.	p_{ex}	Cert.
1	2025-04-19	CORE-Agent	coreagent	+15.6	0.092	8	0.738	0.0304		0.0304		0.449		0.738	
1	2025-04-19	CORE-Agent	coreagent	+13.3	0.145	8	1.000	0.0304		0.0304		0.517		0.702	
2	2025-05-15	HAL Generalist Ag	coreagent	+4.4	1.000	4	1.000	0.0076		0.0076		1.000		1.000	
3	2025-08-06	CORE-Agent	CORE-Agent	+15.6	0.092	4	0.369	0.0034		0.0034		0.225		0.369	
4	2025-08-19	HAL Generalist Ag	coreagent	+15.6	0.065	4	0.262	0.0019		0.0019		0.146		0.262	
4	2025-08-19	HAL Generalist Ag	coreagent	+2.2	1.000	4	1.000	0.0019		0.0019		1.000		1.000	
4	2025-08-19	HAL Generalist Ag	CORE-Agent	+4.4	1.000	4	1.000	0.0019		0.0019		1.000		1.000	
5	2025-08-20	HAL Generalist Ag	HAL Generalist	+2.2	1.000	4	1.000	0.0012		0.0012		1.000		1.000	
6	2025-08-21	CORE-Agent(Deepse	HAL Generalist	+4.4	1.000	3	1.000	0.0008		0.0008		1.000		1.000	
7	2025-10-01	CORE-Agent(Claude	HAL Generalist	+6.7	0.607	2	1.000	0.0006		0.0006		0.944		1.000	
8	2025-11-22	Core Agent	HAL Generalist	+17.8	0.118	2	0.236	0.0005		0.0005		0.125		0.162	
9	2025-12-15	Coreagent(claude-	CORE-Agent(Clau	+31.1	0.002	1	0.002	0.0004		0.0004		0.000	yes	0.001	

Table S16: GAIA: frontier entrants in arrival order. Holder: the most accurate cheaper-or-equal earlier system. p raw: selective record test; g: submissions on the entrant's date; $p = \min(1, g \cdot p_{\text{raw}})$. Levels under alpha-spending and the online fallback, with the certification decision; the last two columns are the composite-null (intersection-union) p-value after the per-date factor and its decision under alpha-spending.

t	Date	Entrant	Holder	Margin	p raw	g	p	Spending	Cert.	Fallback	Cert.	p_{IU}	Cert.	p_{ex}	Cert.
1	2025-04-17	HF Open Deep Rese	HAL Generalist	+0.6	1.000	5	1.000	0.0304		0.0304		1.000		1.000	
2	2025-04-18	HF Open Deep Rese	HF Open Deep Re	+5.5	0.314	1	0.314	0.0076		0.0076		0.270		0.314	
3	2025-04-20	HAL Generalist Ag	HAL Generalist	+4.8	0.438	1	0.438	0.0034		0.0034		0.346		0.361	
4	2025-04-23	HAL Generalist Ag	HAL Generalist	+3.6	0.479	1	0.479	0.0019		0.0019		0.350		0.370	
5	2025-04-25	HAL Generalist Ag	HAL Generalist	+6.1	0.265	2	0.529	0.0012		0.0012		0.401		0.429	
6	2025-08-05	HAL Generalist Ag	HAL Generalist	+0.6	1.000	2	1.000	0.0008		0.0008		1.000		1.000	
7	2025-09-24	HAL Generalist Ag	HAL Generalist	+4.2	0.419	1	0.419	0.0006		0.0006		0.362		0.419	
8	2025-09-26	HAL Generalist Ag	HAL Generalist	+1.2	1.000	1	1.000	0.0005		0.0005		0.799		0.808	
9	2025-09-30	HAL Generalist Ag	HAL Generalist	+6.7	0.185	1	0.185	0.0004		0.0004		0.155		0.185	
10	2025-10-01	HAL Generalist Ag	HAL Generalist	+3.6	0.406	3	1.000	0.0003		0.0003		0.832		0.900	
11	2025-10-14	HAL Generalist Ag	HAL Generalist	+10.3	0.024	2	0.048	0.0003		0.0003		0.036		0.048	

Table S17: Online-Mind2Web: frontier entrants in arrival order. Holder: the most accurate cheaper-or-equal earlier system. p raw: selective record test; g: submissions on the entrant's date; $p = \min(1, g \cdot p_{\text{raw}})$. Levels under alpha-spending and the online fallback, with the certification decision; the last two columns are the composite-null (intersection-union) p-value after the per-date factor and its decision under alpha-spending.

t	Date	Entrant	Holder	Margin	p raw	g	p	Spending	Cert.	Fallback	Cert.	p_{IU}	Cert.	p_{ex}	Cert.
1	2025-05-05	Browser-Use_test(	Browser-Use(gpt	+2.0	0.628	2	1.000	0.0304		0.0304		1.000		1.000	
2	2025-05-09	Browser-Use_test(	Browser-Use(o4-	+14.0	0.000	2	0.000	0.0076	yes	0.0076	yes	0.000	yes	0.000	yes
3	2025-05-18	SeeAct(o3-2025-04	Browser-Use(gpt	+2.7	0.517	1	0.517	0.0034		0.0110		0.427		0.439	
4	2025-08-10	Browser-Use_test(	SeeAct(gemini-2	+0.3	1.000	4	1.000	0.0019		0.0019		1.000		1.000	

t	Date	Entrant	Holder	Margin	p raw	g	p	Spending	Cert.	Fallback	Cert.	p_{IU}	Cert.	p_{ex}	Cert.
4	2025-08-10	Browser-Use_test(	SeeAct(gemini-2	+2.3	0.464	4	1.000	0.0019		0.0019		1.000		1.000	
5	2025-08-11	Browser-Use_test(	SeeAct(gemini-2	+1.3	0.834	2	1.000	0.0012		0.0012		1.000		1.000	
5	2025-08-11	SeeAct(gpt-5-2025	Browser-Use_tes	+1.3	0.810	2	1.000	0.0012		0.0012		1.000		1.000	
6	2025-08-13	SeeAct(gpt-5-2025	SeeAct(gemini-2	+2.7	0.438	2	0.877	0.0008		0.0008		0.693		0.723	
7	2025-10-28	SeeAct(claude-hai	SeeAct(gpt-5-20	+6.0	0.039	4	0.158	0.0006		0.0006		0.115		0.130	
7	2025-10-28	SeeAct(claude-hai	Browser-Use_tes	+3.7	0.272	4	1.000	0.0006		0.0006		0.953		1.000	

Table S18: ScienceAgentBench: frontier entrants in arrival order. Holder: the most accurate cheaper-or-equal earlier system. p raw: selective record test; g: submissions on the entrant's date; $p = \min(1, g \cdot p_{\mathrm{raw}})$. Levels under alpha-spending and the online fallback, with the certification decision; the last two columns are the composite-null (intersection-union) p-value after the per-date factor and its decision under alpha-spending.

t	Date	Entrant	Holder	Margin	p raw	g	p	Spending	Cert.	Fallback	Cert.	p_{IU}	Cert.	p_{ex}	Cert.
1	2025-04-17	SAB Self-Debug Cl	SAB Self-Debug	+4.9	0.383	2	0.767	0.0304		0.0304		0.599		0.767	
2	2025-04-18	sab_selfdebug_gpt	SAB Self-Debug	+8.8	0.049	2	0.098	0.0076		0.0076		0.065		0.098	
3	2025-04-21	SAB Self-Debug o3	sab_selfdebug_g	+8.8	0.136	3	0.408	0.0034		0.0034		0.311		0.408	
3	2025-04-21	SAB Self-Debug o4	sab_selfdebug_g	+2.9	0.678	3	1.000	0.0034		0.0034		1.000		1.000	
3	2025-04-21	SAB Self-Debug o4	SAB Self-Debug	+11.8	0.034	3	0.103	0.0034		0.0034		0.058		0.075	

Table S19: SWE-bench Verified mini: frontier entrants in arrival order. Holder: the most accurate cheaper-or-equal earlier system. p raw: selective record test; g: submissions on the entrant's date; $p = \min(1, g \cdot p_{\mathrm{raw}})$. Levels under alpha-spending and the online fallback, with the certification decision; the last two columns are the composite-null (intersection-union) p-value after the per-date factor and its decision under alpha-spending.

t	Date	Entrant	Holder	Margin	p raw	g	p	Spending	Cert.	Fallback	Cert.	p_{IU}	Cert.	p_{ex}	Cert.
1	2025-04-06	My Agent(o1)	My Agent(gpt-4o)	+12.0	0.227	1	0.227	0.0304		0.0304		0.122		0.148	
2	2025-08-08	SWE-Agent(claude-	My Agent(claude	+4.0	1.000	1	1.000	0.0076		0.0076		0.599		0.632	
3	2025-10-31	SWE-Agent(claude-	SWE-Agent(claud	+14.0	0.065	9	0.589	0.0034		0.0034		0.337		0.589	
3	2025-10-31	SWE-Agent(claude-	My Agent(claude	+16.0	0.028	9	0.256	0.0034		0.0034		0.097		0.160	
3	2025-10-31	SWE-Agent(gpt-5-2	My Agent(claude	+14.0	0.065	9	0.589	0.0034		0.0034		0.337		0.589	
3	2025-10-31	SWE-Agent(claude-	hal_generalist_	+44.0	0.000	9	0.000	0.0034	yes	0.0034	yes	0.000	yes	0.000	yes
4	2025-11-01	SWE-Agent(claude-	SWE-Agent(gpt-5	+8.0	0.501	1	0.501	0.0019		0.0019		0.279		0.316	
5	2025-11-02	SWE-Agent(claude-	SWE-Agent(gpt-5	+4.0	1.000	1	1.000	0.0012		0.0012		0.525		0.571	
6	2025-11-09	HAL Generalist (g	hal_generalist_	+4.0	1.000	4	1.000	0.0008		0.0008		1.000		1.000	
7	2025-11-10	My Agent(gemini/g	hal_generalist_	+20.0	0.008	1	0.008	0.0006		0.0006		0.003		0.005	

Table S20: tau-bench airline: frontier entrants in arrival order. Holder: the most accurate cheaper-or-equal earlier system. p raw: selective record test; g: submissions on the entrant's date; $p = \min(1, g \cdot p_{\mathrm{raw}})$. Levels under alpha-spending and the online fallback, with the certification decision; the last two columns are the composite-null (intersection-union) p-value after the per-date factor and its decision under alpha-spending.

t	Date	Entrant	Holder	Margin	p raw	g	p	Spending	Cert.	Fallback	Cert.	p_{IU}	Cert.	p_{ex}	Cert.
1	2025-04-07	TAU-bench FewShot	TAU-bench FewSh	+12.0	0.319	2	0.639	0.0304		0.0304		0.395		0.448	
2	2025-04-20	TAU-bench FewShot	TAU-bench FewSh	+12.0	0.227	5	1.000	0.0076		0.0076		0.610		0.742	
3	2025-06-25	TAU-bench FewShot	HAL Generalist	+22.0	0.035	1	0.035	0.0034		0.0034		0.020		0.035	
4	2025-08-01	TauBench Few Shot	TAU-bench FewSh	+6.0	0.629	2	1.000	0.0019		0.0019		0.997		1.000	
5	2025-08-05	TauBench FewShot	TAU-bench FewSh	+2.0	1.000	4	1.000	0.0012		0.0012		1.000		1.000	
6	2025-10-13	Taubench ToolCall	HAL Generalist	+6.0	0.664	6	1.000	0.0008		0.0008		1.000		1.000	
7	2025-10-20	Taubench ToolCall	TAU-bench FewSh	+12.0	0.344	6	1.000	0.0006		0.0006		1.000		1.000	

Table S21: USACO: frontier entrants in arrival order. Holder: the most accurate cheaper-or-equal earlier system. p raw: selective record test; g: submissions on the entrant's date; $p = \min(1, g \cdot p_{\mathrm{raw}})$. Levels under alpha-spending and the online fallback, with the certification decision; the last two columns are the composite-null (intersection-union) p-value after the per-date factor and its decision under alpha-spending.

t	Date	Entrant	Holder	Margin	p raw	g	p	Spending	Cert.	Fallback	Cert.	p_{IU}	Cert.	p_{ex}	Cert.
1	2025-04-14	usaco_episodic	usaco_episodic	+5.9	0.011	3	0.033	0.0304		0.0304		0.022	yes	0.026	yes
2	2025-04-15	usaco_episodic	usaco_episodic	+4.9	0.053	1	0.053	0.0076		0.0076		0.043		0.053	
3	2025-04-29	USACO Episodic +	usaco_episodic	+1.3	0.789	1	0.789	0.0034		0.0034		0.636		0.649	
4	2025-05-05	USACO Episodic +	usaco_episodic	+13.0	0.000	1	0.000	0.0019	yes	0.0019	yes	0.000	yes	0.000	yes
5	2025-08-25	USACO Episodic +	USACO Episodic	+11.7	0.000	3	0.000	0.0012	yes	0.0031	yes	0.000	yes	0.000	yes

S3. The frontier winner's curse, system by system

Table S22: Selection-adjusted accuracy of every system on an observed frontier. Pool: the number of systems that cost no more (the system included); a pool of one has no selection and no correction.

Benchmark	System	Pool	Reported	Iterated bootstrap [95% CI]	Conditional [95% CI]
AssistantBench	HAL Generalist Agent (gemini-2.0-fl	1	3.0	3.0 [-2.9, 9.0]	3.0 [-2.9, 9.0]
	HAL Generalist Agent (deepseek-ai/D	5	12.1	10.9 [-0.0, 21.2]	11.6 [-4.1, 23.4]
	HAL Generalist Agent (claude-opus-4	31	15.2	9.8 [-0.6, 20.6]	13.6 [-7.9, 27.5]

Benchmark	System	Pool	Reported	Iterated bootstrap [95% CI]	Conditional [95% CI]
ColBench backend programming	colbench_text_deepseekv3	1	21.0	21.0 [18.5, 23.5]	21.0 [18.5, 23.5]
	colbench_backend_programming colben	2	24.5	24.5 [21.7, 27.1]	24.5 [21.8, 27.2]
	colbench_text_gemmini2flash	3	32.9	32.9 [30.0, 35.8]	32.9 [30.0, 35.8]
	colbench_backend_programming colben	4	33.8	33.3 [30.3, 36.0]	33.6 [29.5, 36.7]
	colbench_text_gpt41	7	35.6	35.3 [32.3, 38.0]	35.6 [32.3, 38.6]
	colbench_backend_programming colben	8	37.2	36.8 [33.8, 39.6]	37.1 [33.4, 40.2]
	colbench_example_agent_gpt41	9	38.6	38.2 [35.3, 40.9]	38.5 [34.7, 41.6]
	colbench_backend_programming colben	10	41.2	41.1 [38.1, 44.1]	41.2 [38.1, 44.3]
	Hal Generalist Agent (Sonnet3.7)	22	43.7	43.2 [40.2, 46.1]	43.6 [39.5, 46.8]
ColBench frontend design	colbench_text_deepseekr1	1	74.9	74.9 [72.8, 77.0]	74.9 [72.8, 77.0]
	colbench_text_deepseekv3	2	75.2	74.7 [72.7, 76.8]	74.1 [67.2, 77.4]
	colbench_text_gemini	3	81.6	81.7 [79.8, 83.6]	81.6 [79.7, 83.6]
	colbench_text_o4minimedium	8	82.9	82.7 [80.9, 84.4]	82.9 [81.1, 84.6]
CORE-Bench hard	HAL Generalist Agent(Deepseek-V3-03	1	8.9	8.9 [0.5, 17.3]	8.9 [0.5, 17.3]
	CORE-Agent	6	11.1	5.1 [-3.4, 12.2]	4.8 [-30.3, 19.5]
	HAL Generalist Agent(o4-mini-low)	8	15.6	10.4 [1.5, 19.3]	13.9 [-5.5, 26.1]
	CORE-Agent(Deepseek-V3.1)	9	20.0	16.0 [6.4, 26.4]	17.0 [-8.8, 31.5]
	HAL Generalist Agent(gpt4.1)	19	22.2	14.9 [5.3, 25.3]	13.3 [-35.2, 33.2]
	Core Agent	23	40.0	40.2 [25.7, 54.6]	40.0 [24.5, 54.5]
	CORE-Agent(Claude-Sonnet-4-5-high)	49	44.4	36.7 [23.3, 50.0]	40.0 [5.0, 58.6]
	Coreagent(claude-opus-4-5)	56	75.6	75.8 [64.7, 89.1]	75.6 [62.9, 88.3]
GAIA	HAL Generalist Agent (deepseek-chat	1	22.4	22.4 [16.0, 28.8]	22.4 [16.0, 28.8]
	HAL Generalist Agent (gemini-2.0-fl	2	32.7	32.9 [26.2, 40.1]	32.7 [25.5, 39.9]
	HAL Generalist Agent (o3-mini-2025-	3	33.9	31.8 [25.0, 37.7]	29.5 [3.9, 40.5]
	HAL Generalist Agent (o3-mini-2025-	5	46.7	46.7 [39.5, 54.1]	46.7 [39.0, 54.3]
	HAL Generalist Agent (gpt-4.1-2025-	7	49.7	47.7 [41.0, 54.3]	48.2 [32.9, 57.2]
	HAL Generalist Agent (o4-mini-2025-	8	54.5	52.9 [45.4, 59.3]	54.2 [43.6, 62.1]
	HAL Generalist Agent (o4-mini-2025-	10	58.2	56.9 [49.2, 63.7]	58.1 [49.2, 65.7]
	HAL Generalist Agent (gpt-5-2025-08	13	59.4	57.0 [50.1, 63.4]	54.7 [27.7, 66.2]
	HAL Generalist Agent (claude-3-7-so	20	64.2	62.6 [55.2, 69.2]	64.0 [54.5, 71.6]
	HAL Generalist Agent (claude-sonnet	23	74.5	74.6 [68.1, 81.4]	74.5 [67.9, 81.2]
Online-Mind2Web	SeeAct(gemini-2.0-flash)	1	26.7	26.7 [21.7, 31.7]	26.7 [21.7, 31.7]
	Browser-Use_test(gemini-2.0-flash)	2	29.0	28.1 [23.2, 32.8]	28.8 [21.8, 34.1]
	SeeAct(claude-haiku-4-5-20251001)	3	32.7	32.1 [26.7, 37.0]	32.6 [26.3, 38.0]
	SeeAct(claude-haiku-4-5-20251001_th	5	36.3	36.3 [30.6, 41.6]	36.3 [30.8, 41.8]
	SeeAct(gemini-2.5-pro-preview-03-25)	6	39.7	39.0 [33.3, 44.3]	39.6 [33.0, 45.2]
	SeeAct(gpt-5-2025-08-07_medium_reas	8	42.3	41.5 [36.0, 47.0]	42.2 [35.5, 47.9]
SciCode	Scicode Zero Shot Agent (gemini-2.0	1	1.5	1.5 [-1.5, 4.6]	1.5 [-1.5, 4.6]
	Scicode Zero Shot Agent (deepseek-a	2	3.1	2.6 [-1.7, 6.0]	2.8 [-3.3, 7.3]
	Scicode Zero Shot Agent (o4-mini-20	3	9.2	9.3 [1.6, 15.5]	9.2 [1.5, 16.3]
ScienceAgentBench	SAB Self-Debug gemini-2-0-flash	1	12.7	12.7 [6.2, 19.2]	12.7 [6.2, 19.2]
	SAB Self-Debug DS-V3	2	15.7	14.9 [7.7, 21.4]	15.6 [7.1, 22.8]
	SAB Self-Debug o4-mini low	3	27.5	27.4 [18.7, 35.4]	27.4 [18.6, 36.2]
	SAB Self-Debug o3 medium	8	33.3	31.7 [22.7, 40.3]	33.2 [22.3, 42.5]
SWE-bench Verified mini	hal_generalist_agent_gemini20flash	1	2.0	2.0 [-1.9, 5.9]	2.0 [-1.9, 5.9]
	My Agent(gemini/gemini-2.0-flash)	4	22.0	22.1 [10.0, 32.0]	22.0 [10.1, 33.6]
	SWE-Agent(claude-3-7-sonnet-2025021	10	54.0	54.3 [40.2, 68.2]	54.0 [40.0, 68.0]
	SWE-Agent(gpt-5-2025-08-07)	13	64.0	63.3 [50.3, 76.3]	64.0 [49.9, 77.4]
	SWE-Agent(claude-sonnet-4-5-2025092	19	68.0	65.7 [53.7, 77.7]	67.7 [51.2, 81.1]
	SWE-Agent(claude-sonnet-4-5-2025092	21	72.0	69.7 [57.7, 81.7]	71.9 [58.1, 84.6]
tau-bench airline	HAL Generalist Agent (gemini-2.0-fl	1	22.0	22.0 [10.4, 33.6]	22.0 [10.4, 33.6]
	Taubench ToolCalling (gemini-2.0-fl	2	28.0	24.3 [13.3, 35.3]	25.1 [-1.3, 40.3]
	TAU-bench FewShot (gemini-2.0-flash)	3	44.0	44.5 [30.9, 56.9]	44.0 [29.7, 57.9]
	Taubench ToolCalling (o4-mini-2025-	6	56.0	54.4 [40.3, 68.3]	55.8 [39.0, 69.9]
	TAU-bench FewShot (o4-mini-2025-04-	13	60.0	55.1 [43.3, 67.3]	57.1 [29.2, 73.4]
	TauBench FewShot High Reasoning (cl	42	62.0	52.6 [40.1, 64.1]	53.6 [5.7, 74.4]
	TauBench Few Shot High Reasoning (c	45	66.0	58.4 [47.0, 69.0]	63.8 [39.2, 79.1]
USACO	usaco_episodic	1	39.1	39.1 [33.6, 44.6]	39.1 [33.6, 44.6]
	usaco_episodic	4	45.0	45.1 [39.2, 50.6]	45.0 [39.4, 50.5]
	USACO Episodic + Semantic (o4-mini-	5	58.0	57.9 [52.4, 63.5]	58.0 [52.5, 63.5]
	USACO Episodic + Semantic (gpt-5-20	8	69.7	69.7 [64.5, 74.9]	69.7 [64.6, 74.9]

S4. Runs left out

Table S23: Runs read from HAL's archives but left out of the audit.

Benchmark	Run	Reason
assistantbench	Assistantbench Browser Agent (gemini-2.0-fl	no logged model calls on 33 of 33 tasks (cost unknown)
assistantbench	Assistantbench Browser Agent (o3-2025-04-16)	outcome coverage 0.00
corebench_hard	Claude Code (claude-opus-4-1)	no logged model calls on 45 of 45 tasks (cost unknown)
corebench_hard	Claude Code (claude-opus-4-5)	no logged model calls on 45 of 45 tasks (cost unknown)
corebench_hard	Claude Code (claude-sonnet-4-20250514)	no logged model calls on 45 of 45 tasks (cost unknown)
corebench_hard	Claude Code (claude-sonnet-4-5-20250929)	no logged model calls on 45 of 45 tasks (cost unknown)
corebench_hard	CORE-Agent	no logged model calls on 45 of 45 tasks (cost unknown)
corebench_hard	HAL Generalist Agent	no logged model calls on 33 of 45 tasks (cost unknown)
gaia	HAL Generalist Agent (o3-2025-04-16)	no logged model calls on 165 of 165 tasks (cost unknown)

Benchmark	Run	Reason
online_mind2web	Browser-Use_test(claude-haiku-4-5-20251001_	outcome coverage 0.88
online_mind2web	Browser-Use_test(claude-haiku-4-5-20251001)	outcome coverage 0.88
online_mind2web	Browser-Use_test(claude-sonnet-4-5-20250929	outcome coverage 0.88
online_mind2web	Browser-Use_test(claude-sonnet-4-5-20250929)	outcome coverage 0.88
swebench_verified_mini	My Agent(o3-mini)	outcome coverage 0.92
swebench_verified_mini	My Agent(o3-mini)	outcome coverage 0.92
swebench_verified_mini	My Agent(o3-2025-04-16)	outcome coverage 0.68
swebench_verified_mini	My Agent(together_ai/deepseek-ai/DeepSeek-R	outcome coverage 0.02
swebench_verified_mini	My Agent(together_ai/deepseek-ai/DeepSeek-V	outcome coverage 0.62
swebench_verified_mini	SWE-Agent(o3-2025-04-16)	outcome coverage 0.94

S5. HAL's double counting of nested spans

HAL's reported token totals sum the usage block of every Weave span, so a provider call logged inside a wrapper span is counted twice. The table gives, per benchmark, how many of the runs read have a ratio of reported to once-counted tokens of exactly two, between one and two, and one.

Table S24: HAL's token total divided by the total with each model call counted once, per run. A ratio of exactly 2 means every provider call was logged inside a wrapper span that carried the same usage block.

Benchmark	Ratio 1	Between 1 and 2	Exactly 2
assistantbench	8	17	8
colbench_backend_programming	17	5	0
colbench_frontend_design	8	6	0
corebench_hard	33	2	26
gaia	12	6	18
scicode	17	11	9
scienceagentbench	13	0	12
swebench_verified_mini	21	0	24
taubench_airline	0	14	33
usaco	1	0	11
All	130	61	141

S6. Price table

Table S25: The price table used for every run (HAL's current table, dollars per million tokens). Cache reads and writes are priced at the cache rate where one is listed and at the input rate otherwise.

Model	Input ($/M)	Output ($/M)	Cache read ($/M)
Meta-Llama-3-1-70B-Instruct-htzs	2.68	3.54	
Meta-Llama-3-1-8B-Instruct-nwxcg	0.3	0.61	
Mistral-large-ygkys	4	12	
Mistral-small-zgjes	1	3	
amazon.nova-lite-v1:0	0.06	0.24	
amazon.nova-micro-v1:0	0.035	0.14	
amazon.nova-pro-v1:0	0.8	3.2	
anthropic/claude-3-5-haiku-20241022	1	5	0.1
anthropic/claude-3-5-sonnet-20240620	3	15	
anthropic/claude-3-5-sonnet-20241022	3	15	
anthropic/claude-3-7-sonnet-20250219	3	15	0.3
anthropic/claude-3.7-sonnet	3	15	
anthropic/claude-3.7-sonnet:thinking	3	15	
anthropic/claude-haiku-3.5	1	5	
anthropic/claude-opus-4	15	75	

Model	Input ($/M)	Output ($/M)	Cache read ($/M)
anthropic/claude-opus-4-20250514	15	75	1.5
anthropic/claude-opus-4-5	5	25	0.5
anthropic/claude-opus-4-7	5	25	0.5
anthropic/claude-opus-4.1	15	75	
anthropic/claude-opus-4.1-20250805	15	75	1.5
anthropic/claude-opus-4.5	5	25	0.5
anthropic/claude-opus-4.7	5	25	0.5
anthropic/claude-sonnet-4	3	15	
anthropic/claude-sonnet-4-20250514	3	15	
anthropic/claude-sonnet-4-5	3	15	0.3
anthropic/claude-sonnet-4-5-20250929	3	15	
bedrock/amazon.nova-lite-v1:0	0.06	0.24	
bedrock/amazon.nova-micro-v1:0	0.035	0.14	
bedrock/amazon.nova-pro-v1:0	0.8	3.2	
bedrock/us.anthropic.claude-3-5-haiku-20241022-v1:0	0.8	4	
bedrock/us.anthropic.claude-3-5-sonnet-20241022-v2:0	3	15	
bedrock/us.anthropic.claude-3-opus-20240229-v1:0	15	75	
bedrock/us.anthropic.claude-3-sonnet-20240229-v1:0	3	15	
bedrock/us.meta.llama3-3-70b-instruct-v1:0	0.72	0.72	
claude-3-5-haiku-20241022	1	5	0.1
claude-3-5-sonnet-20240620	3	15	
claude-3-5-sonnet-20241022	3	15	
claude-3-7-sonnet-20250219	3	15	0.3
claude-haiku-3.5	1	5	
claude-haiku-4-5	1	5	0.1
claude-opus-4	15	75	
claude-opus-4-1-20250805	15	75	1.5
claude-opus-4-20250514	15	75	1.5
claude-opus-4-5	5	25	0.5
claude-opus-4-7	5	25	0.5
claude-opus-4.1	15	75	
claude-opus-4.1-20250805	15	75	1.5
claude-opus-4.5	5	25	0.5
claude-opus-4.7	5	25	0.5
claude-sonnet-4-20250514	3	15	
claude-sonnet-4-5	3	15	0.3
claude-sonnet-4-5-20250929	3	15	0.3
deepseek-ai/DeepSeek-R1	3	7	
deepseek-ai/DeepSeek-V3	1.25	1.25	
deepseek/deepseek-chat-v3-0324	0.18	0.72	
deepseek/deepseek-chat-v3.1	0.27	1.1	
deepseek/deepseek-r1-0528	0.18	0.72	
gemini-2.0-flash	0.1	0.4	0.025
gemini-2.5-flash	0.5	2	0.05
gemini-2.5-pro	1.25	10	0.125
gemini-2.5-pro-preview-03-25	1.25	10	
gemini-3-flash-preview	0.5	3	0.05
gemini-3-pro-preview	2	12	0.2
gemini-3.1-pro-preview	2	12	0.2
gemini-3.5-flash	1.5	9	0.15
gemini/gemini-1.5-flash	0.075	0.3	
gemini/gemini-1.5-flash-latest	0.075	0.3	
gemini/gemini-1.5-pro	1.25	5	
gemini/gemini-2.0-flash	0.1	0.4	0.025
gemini/gemini-2.5-flash	0.5	2	0.05
gemini/gemini-2.5-pro	1.25	10	0.125
gemini/gemini-2.5-pro-preview-03-25	1.25	10	
gemini/gemini-3-flash-preview	0.5	3	0.05

Model	Input ($/M)	Output ($/M)	Cache read ($/M)
gemini/gemini-3-pro-preview	2	12	0.2
gemini/gemini-3.1-pro-preview	2	12	0.2
gemini/gemini-3.5-flash	1.5	9	0.15
google/gemini-1.5-flash	0.075	0.3	
google/gemini-1.5-pro	1.25	5	
google/gemini-2.5-flash	0.5	2	0.05
google/gemini-2.5-pro	1.25	10	0.125
google/gemini-2.5-pro-preview-03-25	1.25	10	
google/gemini-3-flash-preview	0.5	3	0.05
google/gemini-3-pro-preview	2	12	0.2
google/gemini-3.1-pro-preview	2	12	0.2
google/gemini-3.5-flash	1.5	9	0.15
gpt-3.5-turbo	0.5	1.5	
gpt-3.5-turbo-0125	0.5	1.5	
gpt-4-turbo	10	30	
gpt-4-turbo-2024-04-09	10	30	
gpt-4.1-2025-04-14	2	8	0.5
gpt-4.1-mini-2025-04-14	0.4	1.6	
gpt-4.1-nano-2025-04-14	0.1	0.4	
gpt-4.5-preview-2025-02-27	75	150	
gpt-4o	2.5	10	
gpt-4o-2024-05-13	2.5	10	
gpt-4o-2024-08-06	2.5	10	
gpt-4o-2024-11-20	2.5	10	
gpt-4o-mini	0.15	0.6	
gpt-4o-mini-2024-07-18	0.15	0.6	
gpt-5	1.25	10	
gpt-5-2025-08-07	1.25	10	0.125
gpt-5-mini	0.25	2	
gpt-5-nano	0.05	0.4	
gpt-5-pro	15	120	
gpt-5.1	1.25	10	
gpt-5.2	1.75	14	0.18
gpt-5.2-2025-12-11	1.75	14	0.18
gpt-5.2-codex	1.75	14	0.18
gpt-5.2-pro	21	168	
gpt-5.4	2.5	15	0.25
gpt-5.4-mini	0.75	4.5	
gpt-5.4-nano	0.2	1.25	
gpt-5.4-pro	30	180	3
gpt-5.5	5	30	0.5
gpt-5.5-2026-04-23	5	30	0.5
gpt-5.5-pro	30	180	3
meta-llama/Llama-4-Maverick-17B-128E-Instruct-FP8	0.27	0.85	
meta-llama/Meta-Llama-3.1-405B-Instruct	5	15	
meta-llama/Meta-Llama-3.1-70B-Instruct	0.88	0.88	
meta-llama/Meta-Llama-3.1-8B-Instruct	0.18	0.18	
o1-2024-12-17	15	60	
o1-mini-2024-09-12	3	12	
o1-preview-2024-09-12	15	60	
o3-2025-04-16	2	8	0.5
o3-mini-2025-01-31	1.1	4.4	0.55
o4-mini	1.1	4.4	
o4-mini-2025-04-16	1.1	4.4	0.275
openai/gpt-4.1-2025-04-14	2	8	0.5
openai/gpt-4.1-mini-2025-04-14	0.4	1.6	
openai/gpt-4.1-nano-2025-04-14	0.1	0.4	
openai/gpt-4.5-preview-2025-02-27	75	150	

Model	Input ($/M)	Output ($/M)	Cache read ($/M)
openai/gpt-4o-2024-08-06	2.5	10	
openai/gpt-4o-2024-11-20	2.5	10	
openai/gpt-4o-mini	0.15	0.6	
openai/gpt-4o-mini-2024-07-18	0.15	0.6	
openai/gpt-5	1.25	10	
openai/gpt-5-2025-08-07	1.25	10	
openai/gpt-5-mini	0.25	2	
openai/gpt-5-nano	0.05	0.4	
openai/gpt-5-pro	15	120	
openai/gpt-5.1	1.25	10	
openai/gpt-5.2	1.75	14	0.18
openai/gpt-5.2-2025-12-11	1.75	14	0.18
openai/gpt-5.2-codex	1.75	14	0.18
openai/gpt-5.2-pro	21	168	
openai/gpt-5.4	2.5	15	0.25
openai/gpt-5.4-mini	0.75	4.5	
openai/gpt-5.4-nano	0.2	1.25	
openai/gpt-5.4-pro	30	180	3
openai/gpt-5.5	5	30	0.5
openai/gpt-5.5-2026-04-23	5	30	0.5
openai/gpt-5.5-pro	30	180	3
openai/gpt-oss-120b	0.15	0.6	
openai/o1-2024-12-17	15	60	
openai/o1-mini-2024-09-12	3	12	
openai/o1-preview-2024-09-12	15	60	
openai/o3-2025-04-16	2	8	0.5
openai/o3-mini-2025-01-31	1.1	4.4	0.55
openai/o4-mini-2025-04-16	1.1	4.4	0.275
openrouter/anthropic/claude-3-5-haiku	1	5	0.1
openrouter/anthropic/claude-3-7-sonnet-20250219	3	15	0.3
openrouter/anthropic/claude-3-haiku	0.25	1.25	0.03
openrouter/anthropic/claude-3.5-haiku	1	5	0.1
openrouter/anthropic/claude-3.7-sonnet	3	15	
openrouter/anthropic/claude-3.7-sonnet:thinking	3	15	
openrouter/anthropic/claude-opus-4	15	75	
openrouter/anthropic/claude-opus-4-20250514	15	75	
openrouter/anthropic/claude-opus-4.1	15	75	
openrouter/anthropic/claude-opus-4.1-20250805	15	75	
openrouter/anthropic/claude-opus-4.5	5	25	
openrouter/anthropic/claude-sonnet-4	3	15	0.3
openrouter/anthropic/claude-sonnet-4.5	3	15	
openrouter/deepseek/deepseek-chat-v3-0324	0.18	0.72	
openrouter/deepseek/deepseek-chat-v3.1	0.27	1.1	
openrouter/deepseek/deepseek-r1-0528	0.18	0.72	
openrouter/google/gemini-2.0-flash-001	0.1	0.4	0.025
openrouter/google/gemini-2.5-flash	0.3	2.5	0.03
openrouter/google/gemini-2.5-pro	1.25	10	0.125
openrouter/google/gemini-3.1-pro-preview	2	12	0.2
openrouter/google/gemini-3.5-flash	1.5	9	0.15
openrouter/openai/gpt-oss-120b	0.15	0.6	
text-embedding-3-large	0.13	0	
text-embedding-3-small	0.02	0	
together/meta-llama/Meta-Llama-3.1-405B-Instruct-Turbo	3.5	3.5	
together/meta-llama/Meta-Llama-3.1-70B-Instruct	0.88	0.88	
together_ai/deepseek-ai/DeepSeek-R1	3	7	
together_ai/deepseek-ai/DeepSeek-V3	1.25	1.25	
us.anthropic.anthropic.claude-3-sonnet-20240229-v1:0	3	15	
us.anthropic.claude-3-5-haiku-20241022-v1:0	0.8	4	

Model	Input ($/M)	Output ($/M)	Cache read ($/M)
us.anthropic.claude-3-5-sonnet-20240620-v1:0	3	15	
us.anthropic.claude-3-5-sonnet-20241022-v2:0	3	15	
us.anthropic.claude-3-opus-20240229-v1:0	15	75	
us.meta.llama3-3-70b-instruct-v1:0	0.72	0.72	

S7. Sensitivity of the audit to the costing rules

Pricing cache reads and writes at the full input price, and costing a task with no logged model call at zero instead of at the run's mean, are the two rules of the audit that a reader might choose differently.

Table S26: Sensitivity of the audit to two costing rules. Frontier and set: sizes in the audit. Cost ratio: mean cost per task with cache reads and writes priced at the full input price, over the audit's. Frontier moved and set moved: systems that enter or leave the plain frontier or the frontier set under that rule (first pair of columns: full input price for cached tokens; second pair: a task with no logged model call costs nothing instead of the run's mean). Online-Mind2Web has run totals only and is left out of the sensitivity columns. Cost-separated pairs: share of pairs of systems whose cost order the paired step-down of Section 3.3 resolves (boards with per-task costs only).

Benchmark	Sys.	Tasks	Front.	Set	Cost ratio	Front. moved	Set moved	Front. moved	Set moved	Cost-separated pairs
AssistantBench	32	33	3	32	1.01	0	0	0	0	0.78
ColBench backend programming	22	1000	9	14	1.07	0	0	0	0	0.98
ColBench frontend design	14	100	4	7	1.03	0	0	1	0	0.85
CORE-Bench hard	60	45	8	53	1.05	3	1	2	1	0.75
GAIA	36	165	10	22	1.01	2	0	0	0	0.82
SciCode	37	65	3	37	1.02	0	0	0	0	0.92
ScienceAgentBench	25	102	4	21	1.06	0	0	0	0	0.81
SWE-bench Verified mini	39	50	6	26	2.04	3	4	0	0	0.74
tau-bench airline	47	50	7	40	1.04	0	2	0	1	0.82
USACO	12	307	4	5	1.00	0	0	0	0	0.97

S8. All eleven frontiers, and the set sizes

S9. Every archive: HAL's total, our recount, and the run's status

Table S27: Every HAL archive: the benchmark, the archive's name, HAL's reported total cost, the ratio of HAL's token total to the once-counted token total, our once-counted and repriced total cost, and the run's status. The same table, with agent names and the number of duplicate calls, is in `results/hal_reconciliation.csv`.

Benchmark	Archive	Reported	Ratio	Recounted	Status
	scicode_hal_generalist_agent_gem				skipped: results member cannot
	seeact_gpt-5_reasoning_effort_hi				skipped: 12.8 GB upload cannot
	usaco_hal_generalist_agent_gpt41				skipped: results member (3.9 G
	usaco_usaco_episodic__semantic_c				skipped: MemberTooLarge: resul
	usaco_usaco_episodic__semantic_g				skipped: MemberTooLarge: resul
assistantbench	assistantbench_assistantbench_br	56.00	3.92	14.32	kept
assistantbench	assistantbench_assistantbench_br	16.13	1.00	16.13	kept
assistantbench	assistantbench_assistantbench_br	14.88	1.00	14.88	kept
assistantbench	assistantbench_assistantbench_br	385.43	3.92	98.45	kept
assistantbench	assistantbench_assistantbench_br	779.72	2.34	333.11	kept
assistantbench	assistantbench_assistantbench_br	75.60	3.99	18.96	kept
assistantbench	assistantbench_assistantbench_br	0.00	3.98	24.93	kept
assistantbench	assistantbench_assistantbench_br	5.30	3.97	1.34	kept
assistantbench	assistantbench_assistantbench_br	12.66	3.81	3.32	kept
assistantbench	assistantbench_assistantbench_br	86.45	2.37	36.53	kept
assistantbench	assistantbench_assistantbench_br	0.00		0.00	dropped: no logged model calls
assistantbench	assistantbench_assistantbench_br	2.18	2.37	0.92	kept
assistantbench	assistantbench_assistantbench_br	14.15	3.77	2.76	kept
assistantbench	assistantbench_assistantbench_br	41.84	3.72	9.68	kept
assistantbench	assistantbench_assistantbench_br	41.69	3.79	9.75	kept
assistantbench	assistantbench_assistantbench_br	60.24	3.97	2.20	dropped: outcome coverage 0.00
assistantbench	assistantbench_assistantbench_br	75.77	3.95	2.93	kept
assistantbench	assistantbench_assistantbench_br	12.50	3.70	3.10	kept
assistantbench	assistantbench_assistantbench_br	16.39	3.86	3.95	kept
assistantbench	assistantbench_assistantbench_br	9.22	3.93	2.04	kept

Benchmark	Archive	Reported	Ratio	Recounted	Status
assistantbench	assistantbench_hal_generalist_ag	9.71	1.00	9.71	kept
assistantbench	assistantbench_hal_generalist_ag	10.56	1.00	10.56	kept
assistantbench	assistantbench_hal_generalist_ag	9.81	1.00	9.81	kept
assistantbench	assistantbench_hal_generalist_ag	108.47	1.00	108.46	kept
assistantbench	assistantbench_hal_generalist_ag	112.61	1.00	112.61	kept
assistantbench	assistantbench_hal_generalist_ag	145.01	1.00	145.01	kept
assistantbench	assistantbench_hal_generalist_ag	24.44	2.00	12.22	kept
assistantbench	assistantbench_hal_generalist_ag	3.98	2.00	1.99	kept
assistantbench	assistantbench_hal_generalist_ag	1.40	2.00	0.70	kept
assistantbench	assistantbench_hal_generalist_ag	10.83	2.00	3.43	kept
assistantbench	assistantbench_hal_generalist_ag	30.93	2.00	2.41	kept
assistantbench	assistantbench_hal_generalist_ag	5.90	2.00	2.64	kept
assistantbench	assistantbench_hal_generalist_ag	8.84	2.00	4.18	kept
assistantbench	assistantbench_hal_generalist_ag	2.11	2.00	0.96	kept
colbench_backe	colbench_backend_programming_col	23.63	1.00	23.63	kept
colbench_backe	colbench_backend_programming_col	40.60	1.00	40.60	kept
colbench_backe	colbench_backend_programming_col	35.15	1.00	35.15	kept
colbench_backe	colbench_backend_programming_col	2.10	1.00	2.10	kept
colbench_backe	colbench_backend_programming_col	6.01	1.00	6.01	kept
colbench_backe	colbench_backend_programming_col	52.55	1.00	52.55	kept
colbench_backe	colbench_backend_programming_col	106.60	1.00	26.19	kept
colbench_backe	colbench_backend_programming_col	31.38	1.00	31.38	kept
colbench_backe	colbench_backend_programming_col	14.73	1.00	14.72	kept
colbench_backe	colbench_backend_programming_col	204.80	1.92	105.94	kept
colbench_backe	colbench_backend_programming_col	53.26	1.89	29.85	kept
colbench_backe	colbench_backend_programming_col	15.99	1.00	15.99	kept
colbench_backe	colbench_backend_programming_col	143.33	1.89	55.63	kept
colbench_backe	colbench_backend_programming_col	1,299.27	1.96	106.11	kept
colbench_backe	colbench_backend_programming_col	138.13	1.92	63.29	kept
colbench_backe	colbench_backend_programming_col	4.14	1.00	4.14	kept
colbench_backe	colbench_backend_programming_col	1.49	1.00	1.49	kept
colbench_backe	colbench_backend_programming_col	5.88	1.00	5.88	kept
colbench_backe	colbench_backend_programming_col	13.99	1.00	13.99	kept
colbench_backe	colbench_backend_programming_col	8.55	1.00	8.55	kept
colbench_backe	colbench_backend_programming_col	33.77	1.00	33.77	kept
colbench_backe	colbench_backend_programming_hal	202.35	1.00	202.35	kept
colbench_front	colbench_frontend_design_colbenc	33.28	1.92	17.24	kept
colbench_front	colbench_frontend_design_colbenc	11.08	1.93	6.03	kept
colbench_front	colbench_frontend_design_colbenc	15.71	1.91	6.52	kept
colbench_front	colbench_frontend_design_colbenc	20.62	1.92	9.70	kept
colbench_front	colbench_frontend_design_colbenc	153.99	1.95	13.16	kept
colbench_front	colbench_frontend_design_colbenc	29.43	1.90	14.41	kept
colbench_front	colbench_frontend_design_colbenc	0.85	1.00	0.85	kept
colbench_front	colbench_frontend_design_colbenc	0.84	1.00	0.84	kept
colbench_front	colbench_frontend_design_colbenc	2.66	1.00	2.66	kept
colbench_front	colbench_frontend_design_colbenc	14.57	1.00	14.23	kept
colbench_front	colbench_frontend_design_colbenc	6.21	1.00	6.08	kept
colbench_front	colbench_frontend_design_colbenc	12.76	1.00	12.35	kept
colbench_front	colbench_frontend_design_colbenc	71.30	1.00	71.30	kept
colbench_front	colbench_frontend_design_halgene	104.84	1.00	104.84	kept
corebench_hard	corebench_hard_claude_code_claud	331.79		0.00	dropped: no logged model calls
corebench_hard	corebench_hard_claude_code_claud	87.16		0.00	dropped: no logged model calls
corebench_hard	corebench_hard_claude_code_claud	65.58		0.00	dropped: no logged model calls
corebench_hard	corebench_hard_claude_code_claud	68.33		0.00	dropped: no logged model calls
corebench_hard	corebench_hard_core_agent_176384	0.01	2.00	15.22	kept
corebench_hard	corebench_hard_core_agent_opus_4	168.99	1.00	168.97	kept
corebench_hard	corebench_hard_core_agent_opus_4	152.66	1.00	152.56	kept
corebench_hard	corebench_hard_coreagent_1744752	107.36	2.00	23.73	kept
corebench_hard	corebench_hard_coreagent_1744839	21.86	2.00	7.15	kept
corebench_hard	corebench_hard_coreagent_1744854	25.26	2.00	12.63	kept
corebench_hard	corebench_hard_coreagent_1744856	12.46	2.00	6.23	kept
corebench_hard	corebench_hard_coreagent_1744922	73.04	1.00	72.99	kept
corebench_hard	corebench_hard_coreagent_1744922	182.34	2.00	91.17	kept
corebench_hard	corebench_hard_coreagent_1744922	25.06	2.00	8.63	kept
corebench_hard	corebench_hard_coreagent_1744922	127.57	2.00	63.78	kept
corebench_hard	corebench_hard_coreagent_1745046	31.79	2.00	12.00	kept
corebench_hard	corebench_hard_coreagent_1745075	61.35	2.00	24.57	kept
corebench_hard	corebench_hard_coreagent_1745118	602.37	2.00	31.16	kept
corebench_hard	corebench_hard_coreagent_1745118	75.19	1.00	75.19	kept
corebench_hard	corebench_hard_coreagent_1745258	72.47	1.00	72.40	kept
corebench_hard	corebench_hard_coreagent_1754492	4.21	1.00	4.21	kept
corebench_hard	corebench_hard_coreagent_1754492	412.42	1.00	412.40	kept
corebench_hard	corebench_hard_coreagent_1754539	4.21	1.00	4.20	kept
corebench_hard	corebench_hard_coreagent_1754539	509.95	1.00	509.89	kept
corebench_hard	corebench_hard_coreagent_1754599	31.76		0.00	dropped: no logged model calls
corebench_hard	corebench_hard_coreagentclaudeha	0.00	1.00	43.93	kept
corebench_hard	corebench_hard_coreagentclaudeop	273.74	1.00	273.69	kept
corebench_hard	corebench_hard_coreagentclaudeso	97.15	1.00	97.14	kept
corebench_hard	corebench_hard_coreagentclaudeso	92.34	1.00	92.33	kept
corebench_hard	corebench_hard_coreagentclaudeso	50.27	1.00	50.18	kept
corebench_hard	corebench_hard_coreagentclaudeso	100.48	1.00	100.42	kept

Benchmark	Archive	Reported	Ratio	Recounted	Status
corebench_hard	corebench_hard_coreagentdeepseek	35.43	1.00	34.66	kept
corebench_hard	corebench_hard_coreagentdeepseek	5.91	1.00	5.87	kept
corebench_hard	corebench_hard_hal_generalist_ag	54.61	2.00	13.99	kept
corebench_hard	corebench_hard_hal_generalist_ag	18.50	2.00	9.25	kept
corebench_hard	corebench_hard_hal_generalist_ag	0.16	1.00	0.16	dropped: no logged model calls
corebench_hard	corebench_hard_hal_generalist_ag	17.68	2.00	8.84	kept
corebench_hard	corebench_hard_hal_generalist_ag	21.80	2.00	9.38	kept
corebench_hard	corebench_hard_hal_generalist_ag	47.10	2.00	23.55	kept
corebench_hard	corebench_hard_hal_generalist_ag	30.38	2.00	15.19	kept
corebench_hard	corebench_hard_hal_generalist_ag	50.28	2.00	22.69	kept
corebench_hard	corebench_hard_hal_generalist_ag	123.50	2.00	8.57	kept
corebench_hard	corebench_hard_hal_generalist_ag	36.67	1.00	36.67	kept
corebench_hard	corebench_hard_hal_generalist_ag	63.21	1.00	63.21	kept
corebench_hard	corebench_hard_hal_generalist_ag	51.59	1.00	51.59	kept
corebench_hard	corebench_hard_hal_generalist_ag	54.37	1.00	54.37	kept
corebench_hard	corebench_hard_hal_generalist_ag	2.79	1.00	2.79	kept
corebench_hard	corebench_hard_hal_generalist_ag	375.11	1.00	375.11	kept
corebench_hard	corebench_hard_hal_generalist_ag	2.05	1.00	2.05	kept
corebench_hard	corebench_hard_hal_generalist_ag	358.47	1.00	358.47	kept
corebench_hard	corebench_hard_hal_generalist_ag	0.01	2.00	17.38	kept
corebench_hard	corebench_hard_hal_generalist_ag	127.41	1.00	127.40	kept
corebench_hard	corebench_hard_hal_generalist_ag	112.38	1.00	112.37	kept
corebench_hard	corebench_hard_hal_generalist_ag	2.55	1.00	2.55	kept
corebench_hard	corebench_hard_hal_generalist_ag	14.37	1.00	13.87	kept
corebench_hard	corebench_hard_hal_generalist_ag	0.76	1.00	0.76	kept
corebench_hard	corebench_hard_hal_generalist_ag	7.06	2.00	3.53	kept
corebench_hard	corebench_hard_hal_generalist_ag	58.32	2.00	13.32	kept
corebench_hard	corebench_hard_hal_generalist_ag	29.75	2.00	14.07	kept
corebench_hard	corebench_hard_hal_generalist_ag	441.70	2.00	20.68	kept
corebench_hard	corebench_hard_hal_generalist_ag	45.37	2.00	16.44	kept
corebench_hard	corebench_hard_hal_generalist_ag	22.50	2.00	5.45	kept
corebench_hard	corebench_hard_hal_generalist_ag	56.64	1.00	56.62	kept
corebench_hard	corebench_hard_hal_generalist_ag	66.15	1.00	66.11	kept
corebench_hard	corebench_hard_halgeneralistagen	85.19	1.00	85.13	kept
corebench_hard	corebench_hard_halgeneralistagen	87.77	1.00	87.75	kept
gaia	gaia_hal_generalist_agent_claude	130.68	1.00	130.52	kept
gaia	gaia_hal_generalist_agent_claude	122.49	1.00	122.39	kept
gaia	gaia_hal_generalist_agent_claude	0.29	1.00	130.66	kept
gaia	gaia_hal_generalist_agent_claude	562.24	1.00	562.14	kept
gaia	gaia_hal_generalist_agent_claude	641.86	1.00	641.73	kept
gaia	gaia_hal_generalist_agent_claude	272.76	1.00	272.73	kept
gaia	gaia_hal_generalist_agent_claude	665.89	1.00	665.75	kept
gaia	gaia_hal_generalist_agent_claude	179.86	1.00	179.61	kept
gaia	gaia_hal_generalist_agent_claude	187.37	1.00	187.06	kept
gaia	gaia_hal_generalist_agent_claude	169.02	1.00	168.83	kept
gaia	gaia_hal_generalist_agent_deepse	73.19	2.00	36.60	kept
gaia	gaia_hal_generalist_agent_deepse	29.27	2.00	14.63	kept
gaia	gaia_hal_generalist_agent_deepse	5.52	1.00	0.93	kept
gaia	gaia_hal_generalist_agent_gemini	7.80	2.00	3.90	kept
gaia	gaia_hal_generalist_agent_gpt412	74.19	2.00	20.03	kept
gaia	gaia_hal_generalist_agent_gpt520	104.75	2.00	36.11	kept
gaia	gaia_hal_generalist_agent_o32025	2,828.54		0.00	dropped: no logged model calls
gaia	gaia_hal_generalist_agent_o3mini	13.19	2.00	5.28	kept
gaia	gaia_hal_generalist_agent_o3mini	30.37	2.00	14.18	kept
gaia	gaia_hal_generalist_agent_o4mini	59.39	2.00	23.85	kept
gaia	gaia_hal_generalist_agent_o4mini	73.26	2.00	28.94	kept
gaia	gaia_hf_open_deep_research_claud	415.15	1.00	415.15	kept
gaia	gaia_hf_open_deep_research_claud	113.65	1.00	113.65	kept
gaia	gaia_hf_open_deep_research_claud	1,306.85	1.00	1,306.85	kept
gaia	gaia_hf_open_deep_research_claud	1,473.64	1.00	1,473.64	kept
gaia	gaia_hf_open_deep_research_claud	1,686.07	1.00	1,686.07	kept
gaia	gaia_hf_open_deep_research_claud	535.00	1.01	532.54	kept
gaia	gaia_hf_open_deep_research_claud	451.84	1.01	449.42	kept
gaia	gaia_hf_open_deep_research_deeps	143.08	2.00	71.54	kept
gaia	gaia_hf_open_deep_research_deeps	76.64	2.00	38.32	kept
gaia	gaia_hf_open_deep_research_gemin	18.82	2.00	9.41	kept
gaia	gaia_hf_open_deep_research_gpt41	109.88	2.00	31.13	kept
gaia	gaia_hf_open_deep_research_gpt52	359.83	2.00	168.92	kept
gaia	gaia_hf_open_deep_research_o3202	681.97	2.00	46.66	kept
gaia	gaia_hf_open_deep_research_o3min	57.24	2.00	25.31	kept
gaia	gaia_hf_open_deep_research_o4min	184.87	2.00	79.56	kept
gaia	gaia_hf_open_deep_research_o4min	80.80	2.00	32.03	kept
online_mind2we	browser-use_DeepSeek-R1	280.93		0.00	kept
online_mind2we	browser-use_DeepSeek-V3	214.74		0.00	kept
online_mind2we	browser-use_claude-3-7-sonnet-20	926.48		0.00	kept
online_mind2we	browser-use_claude-3-7-sonnet-20	1,151.88		0.00	kept
online_mind2we	browser-use_claude-haiku-4-5-202	598.68		0.00	dropped: outcome coverage 0.88
online_mind2we	browser-use_claude-haiku-4-5-202	617.52		0.00	dropped: outcome coverage 0.88
online_mind2we	browser-use_claude-sonnet-4-2025	1,577.26		0.00	kept
online_mind2we	browser-use_claude-sonnet-4-2025	1,609.92		0.00	kept
online_mind2we	browser-use_claude-sonnet-4-5-20	1,659.81		0.00	dropped: outcome coverage 0.88

Benchmark	Archive	Reported	Ratio	Recounted	Status
online_mind2we	browser-use_claude-sonnet-4-5-20	1,669.28		0.00	dropped: outcome coverage 0.88
online_mind2we	browser-use_gemini-2.0-flash	18.01		0.00	kept
online_mind2we	browser-use_gemini-2.5-pro-previ	331.37		0.00	kept
online_mind2we	browser-use_gpt-4.1-2025-04-14	236.62		0.00	kept
online_mind2we	browser-use_gpt-5_reasoning_effo	1,069.59		0.00	kept
online_mind2we	browser-use_gpt-5_reasoning_effo	736.31		0.00	kept
online_mind2we	browser-use_gpt-5_reasoning_effo	347.23		0.00	kept
online_mind2we	browser-use_o3-2025-04-16	1,857.94		0.00	kept
online_mind2we	browser-use_o4-mini_reasoning_ef	297.93		0.00	kept
online_mind2we	browser-use_o4-mini_reasoning_ef	201.44		0.00	kept
online_mind2we	seeact_claude-3-7-sonnet-20250219	291.97		0.00	kept
online_mind2we	seeact_claude-3-7-sonnet-2025021	367.51		0.00	kept
online_mind2we	seeact_claude-haiku-4-5-20251001	66.78		0.00	kept
online_mind2we	seeact_claude-haiku-4-5-20251001	89.17		0.00	kept
online_mind2we	seeact_claude-sonnet-4-20250514	246.18		0.00	kept
online_mind2we	seeact_claude-sonnet-4-20250514_	326.41		0.00	kept
online_mind2we	seeact_claude-sonnet-4-5-20250929	204.86		0.00	kept
online_mind2we	seeact_claude-sonnet-4-5-2025092	293.15		0.00	kept
online_mind2we	seeact_gemini-2.0-flash	5.03		0.00	kept
online_mind2we	seeact_gemini-2.5-pro-preview-03	153.61		0.00	kept
online_mind2we	seeact_gpt-4.1-2025-04-14	271.24		0.00	kept
online_mind2we	seeact_gpt-5_reasoning_effort_hi	342.93		0.00	kept
online_mind2we	seeact_gpt-5_reasoning_effort_me	171.07		0.00	kept
online_mind2we	seeact_gpt-5_reasoning_effort_mi	78.86		0.00	kept
online_mind2we	seeact_o3-2025-04-16	1,293.69		0.00	kept
online_mind2we	seeact_o4-mini_reasoning_effort_	228.98		0.00	kept
online_mind2we	seeact_o4-mini_reasoning_effort_	162.36		0.00	kept
scicode	scicode_hal_generalist_agent_cla	60.40	1.00	60.40	kept
scicode	scicode_hal_generalist_agent_cla	188.15	1.00	188.15	kept
scicode	scicode_hal_generalist_agent_dee	486.78	2.00	243.39	kept
scicode	scicode_hal_generalist_agent_dee	219.36	2.00	109.68	kept
scicode	scicode_hal_generalist_agent_gpt	73.87	2.00	23.32	kept
scicode	scicode_hal_generalist_agent_o32	291.50	2.00	23.06	kept
scicode	scicode_hal_generalist_agent_o32	334.88	2.00	24.83	kept
scicode	scicode_hal_generalist_agent_o4m	87.69	2.00	37.47	kept
scicode	scicode_hal_generalist_agent_o4m	92.10	2.00	39.89	kept
scicode	scicode_hal_generalist_agent_o4m	165.90	2.00	66.87	kept
scicode	scicode_scicode-tool_calling_age	507.73	1.00	507.73	kept
scicode	scicode_scicode_tool_calling_age	191.41	1.98	97.46	kept
scicode	scicode_scicode_tool_calling_age	204.37	1.98	103.96	kept
scicode	scicode_scicode_tool_calling_age	232.36	1.99	117.48	kept
scicode	scicode_scicode_tool_calling_age	625.13	1.00	625.13	kept
scicode	scicode_scicode_tool_calling_age	593.35	1.00	593.35	kept
scicode	scicode_scicode_tool_calling_age	104.29	1.00	104.29	kept
scicode	scicode_scicode_tool_calling_age	0.00	1.00	118.14	kept
scicode	scicode_scicode_tool_calling_age	52.11	1.98	26.29	kept
scicode	scicode_scicode_tool_calling_age	273.98	1.98	139.21	kept
scicode	scicode_scicode_tool_calling_age	5.23	1.98	2.65	kept
scicode	scicode_scicode_tool_calling_age	69.39	1.99	25.59	kept
scicode	scicode_scicode_tool_calling_age	193.52	2.00	93.70	kept
scicode	scicode_scicode_tool_calling_age	555.57	1.97	46.93	kept
scicode	scicode_scicode_tool_calling_age	55.86	1.97	25.91	kept
scicode	scicode_scicode_tool_calling_age	66.20	1.97	32.14	kept
scicode	scicode_scicode_tool_calling_age	46.30	1.98	20.83	kept
scicode	scicode_scicode_zero_shot_agent_	5.10	1.00	5.10	kept
scicode	scicode_scicode_zero_shot_agent_	4.99	1.00	4.99	kept
scicode	scicode_scicode_zero_shot_agent_	9.18	1.00	9.18	kept
scicode	scicode_scicode_zero_shot_agent_	0.79	1.00	0.79	kept
scicode	scicode_scicode_zero_shot_agent_	0.12	1.00	0.12	kept
scicode	scicode_scicode_zero_shot_agent_	2.82	1.00	2.52	kept
scicode	scicode_scicode_zero_shot_agent_	30.16	1.00	5.79	kept
scicode	scicode_scicode_zero_shot_agent_	2.72	1.00	2.70	kept
scicode	scicode_scicode_zero_shot_agent_	5.37	1.00	5.34	kept
scicode	scicode_scicode_zero_shot_agent_	1.74	1.00	1.70	kept
scienceagentbe	scienceagentbench_hal_generalist	41.22	1.00	41.22	kept
scienceagentbe	scienceagentbench_hal_generalist	48.28	1.00	48.28	kept
scienceagentbe	scienceagentbench_hal_generalist	55.73	2.00	27.87	kept
scienceagentbe	scienceagentbench_hal_generalist	68.95	2.00	23.63	kept
scienceagentbe	scienceagentbench_hal_generalist	155.42	2.00	14.12	kept
scienceagentbe	scienceagentbench_hal_generalist	76.30	2.00	33.80	kept
scienceagentbe	scienceagentbench_hal_generalist	77.32	2.00	34.32	kept
scienceagentbe	scienceagentbench_sab_selfdebug_	7.12	1.00	7.12	kept
scienceagentbe	scienceagentbench_sab_selfdebug_	11.74	1.00	11.74	kept
scienceagentbe	scienceagentbench_sab_selfdebug_	6.56	1.00	6.56	kept
scienceagentbe	scienceagentbench_sab_selfdebug_	2.66	1.00	2.66	kept
scienceagentbe	scienceagentbench_sab_selfdebug_	3.41	1.00	3.41	kept
scienceagentbe	scienceagentbench_sab_selfdebug_	33.37	1.00	33.37	kept
scienceagentbe	scienceagentbench_sab_selfdebug_	33.75	1.00	33.75	kept
scienceagentbe	scienceagentbench_sab_selfdebug_	7.39	1.00	7.39	kept
scienceagentbe	scienceagentbench_sab_selfdebug_	7.47	1.00	7.47	kept
scienceagentbe	scienceagentbench_sab_selfdebug_	18.24	2.00	9.12	kept

Benchmark	Archive	Reported	Ratio	Recounted	Status
scienceagentbe	scienceagentbench_sab_selfdebug_	2.09	2.00	1.05	kept
scienceagentbe	scienceagentbench_sab_selfdebug_	0.19	1.00	0.19	kept
scienceagentbe	scienceagentbench_sab_selfdebug_	11.20	1.00	11.20	kept
scienceagentbe	scienceagentbench_sab_selfdebug_	18.26	2.00	9.12	kept
scienceagentbe	scienceagentbench_sab_selfdebug_	7.42	2.00	3.49	kept
scienceagentbe	scienceagentbench_sab_selfdebug_	58.44	2.00	5.83	kept
scienceagentbe	scienceagentbench_sab_selfdebug_	11.18	2.00	5.59	kept
scienceagentbe	scienceagentbench_sab_selfdebug_	3.95	2.00	1.94	kept
swebench_verif	swebench_verified_mini_1743738168	350.72	2.00	175.36	kept
swebench_verif	swebench_verified_mini_1743879454	113.33	2.00	33.31	dropped: outcome coverage 0.92
swebench_verif	swebench_verified_mini_1743903907	447.78	2.00	223.89	kept
swebench_verif	swebench_verified_mini_1744145062	182.30	2.00	59.37	kept
swebench_verif	swebench_verified_mini_1744313672	64.18	2.00	18.28	dropped: outcome coverage 0.92
swebench_verif	swebench_verified_mini_1744335342	34.07	1.00	34.07	kept
swebench_verif	swebench_verified_mini_1744384313	489.29	1.00	39.69	kept
swebench_verif	swebench_verified_mini_1744397462	33.34	1.00	33.34	kept
swebench_verif	swebench_verified_mini_1744505272	4.72	1.00	4.72	kept
swebench_verif	swebench_verified_mini_hal_gener	117.43	1.00	117.43	kept
swebench_verif	swebench_verified_mini_hal_gener	146.71	2.00	73.35	kept
swebench_verif	swebench_verified_mini_hal_gener	30.17	2.00	15.08	kept
swebench_verif	swebench_verified_mini_hal_gener	7.33	2.00	3.67	kept
swebench_verif	swebench_verified_mini_hal_gener	51.80	2.00	16.18	kept
swebench_verif	swebench_verified_mini_hal_gener	149.59	2.00	74.80	kept
swebench_verif	swebench_verified_mini_hal_gener	8.09	2.00	3.76	kept
swebench_verif	swebench_verified_mini_hal_gener	11.08	2.00	3.87	kept
swebench_verif	swebench_verified_mini_hal_gener	32.02	2.00	11.45	kept
swebench_verif	swebench_verified_mini_hal_gener	87.03	2.00	19.89	kept
swebench_verif	swebench_verified_mini_hal_gener	76.81	1.00	76.81	kept
swebench_verif	swebench_verified_mini_hal_gener	65.31	1.00	65.31	kept
swebench_verif	swebench_verified_mini_hal_gener	87.22	1.00	87.22	kept
swebench_verif	swebench_verified_mini_hal_gener	208.56	1.00	208.56	kept
swebench_verif	swebench_verified_mini_hal_gener	399.93	1.00	399.93	kept
swebench_verif	swebench_verified_mini_hal_gener	477.65	1.00	477.65	kept
swebench_verif	swebench_verified_mini_hal_gener	403.42	1.00	403.42	kept
swebench_verif	swebench_verified_mini_hal_gener	382.39	1.00	382.39	kept
swebench_verif	swebench_verified_mini_hal_gener	95.97	1.00	95.97	kept
swebench_verif	swebench_verified_mini_hal_gener	128.19	1.00	128.19	kept
swebench_verif	swebench_verified_mini_hal_gener	45.09	2.00	22.54	kept
swebench_verif	swebench_verified_mini_hal_gener	2,928.55	2.00	94.18	kept
swebench_verif	swebench_verified_mini_my_agentc	105.97	1.00	105.97	kept
swebench_verif	swebench_verified_mini_my_agentg	126.50	2.00	63.25	kept
swebench_verif	swebench_verified_mini_my_agento	421.79	2.00	37.96	dropped: outcome coverage 0.68
swebench_verif	swebench_verified_mini_my_agento	234.69	2.00	117.35	kept
swebench_verif	swebench_verified_mini_my_agento	227.20	2.00	113.60	kept
swebench_verif	swebench_verified_mini_my_agentt	4.16	2.00	2.08	dropped: outcome coverage 0.02
swebench_verif	swebench_verified_mini_my_agentt	11.77	2.00	5.89	dropped: outcome coverage 0.62
swebench_verif	swebench_verified_mini_sweagentc	131.15	1.00	131.15	kept
swebench_verif	swebench_verified_mini_sweagentc	1,789.67	1.00	140.53	kept
swebench_verif	swebench_verified_mini_sweagentc	170.13	1.00	170.13	kept
swebench_verif	swebench_verified_mini_sweagentc	78.67	1.00	78.67	kept
swebench_verif	swebench_verified_mini_sweagentc	75.43	1.00	75.43	kept
swebench_verif	swebench_verified_mini_sweagentg	102.00	2.00	51.00	kept
swebench_verif	swebench_verified_mini_sweagento	182.61	2.00	91.31	dropped: outcome coverage 0.94
taubench_airli	taubench_airline_1743961943	46.47	2.00	23.23	kept
taubench_airli	taubench_airline_1743969279	20.56	2.00	7.01	kept
taubench_airli	taubench_airline_1743970420	20.05	2.00	6.40	kept
taubench_airli	taubench_airline_1743994890	36.45	1.03	35.62	kept
taubench_airli	taubench_airline_1744000536	37.23	1.03	36.29	kept
taubench_airli	taubench_airline_hal_generalist_	42.11	1.03	41.18	kept
taubench_airli	taubench_airline_hal_generalist_	34.58	1.03	33.84	kept
taubench_airli	taubench_airline_hal_generalist_	30.18	2.00	15.09	kept
taubench_airli	taubench_airline_hal_generalist_	10.73	2.00	5.36	kept
taubench_airli	taubench_airline_hal_generalist_	2.00	2.00	1.00	kept
taubench_airli	taubench_airline_hal_generalist_	17.85	2.00	5.76	kept
taubench_airli	taubench_airline_hal_generalist_	221.59	2.00	13.15	kept
taubench_airli	taubench_airline_hal_generalist_	20.57	2.00	8.69	kept
taubench_airli	taubench_airline_hal_generalist_	20.16	2.00	8.65	kept
taubench_airli	taubench_airline_hal_generalist_	150.15	1.03	149.36	kept
taubench_airli	taubench_airline_hal_generalist_	52.78	2.00	22.37	kept
taubench_airli	taubench_airline_hal_generalist_	140.28	1.03	139.64	kept
taubench_airli	taubench_airline_hal_generalist_	150.29	1.03	149.61	kept
taubench_airli	taubench_airline_hal_generalist_	62.11	2.00	25.85	kept
taubench_airli	taubench_airline_hal_generalist_	180.49	1.03	179.67	kept
taubench_airli	taubench_airline_taubench_few_sh	363.30	2.00	181.65	kept
taubench_airli	taubench_airline_taubench_few_sh	313.83	2.00	156.92	kept
taubench_airli	taubench_airline_taubench_fewsho	31.75	2.00	15.87	kept
taubench_airli	taubench_airline_taubench_fewsho	30.60	2.00	15.30	kept
taubench_airli	taubench_airline_taubench_fewsho	4.44	2.00	2.22	kept
taubench_airli	taubench_airline_taubench_fewsho	42.58	2.00	10.63	kept
taubench_airli	taubench_airline_taubench_fewsho	298.58	2.00	149.29	kept
taubench_airli	taubench_airline_taubench_fewsho	35.49	2.00	12.94	kept

Benchmark	Archive	Reported	Ratio	Recounted	Status
taubench_airli	taubench_airline_taubench_fewsho	33.05	2.00	12.42	kept
taubench_airli	taubench_airline_taubench_fewsho	294.17	2.00	147.08	kept
taubench_airli	taubench_airline_taubench_fewsho	237.93	2.00	8.91	kept
taubench_airli	taubench_airline_taubench_fewsho	170.28	2.00	12.08	kept
taubench_airli	taubench_airline_taubench_fewsho	32.23	2.00	12.14	kept
taubench_airli	taubench_airline_taubench_fewsho	18.92	2.00	7.27	kept
taubench_airli	taubench_airline_taubench_fewsho	18.81	2.00	7.45	kept
taubench_airli	taubench_airline_taubench_toolca	15.45	1.05	16.19	kept
taubench_airli	taubench_airline_taubench_toolca	31.94	2.00	16.90	kept
taubench_airli	taubench_airline_taubench_toolca	69.78	1.08	70.73	kept
taubench_airli	taubench_airline_taubench_toolca	149.98	2.00	75.88	kept
taubench_airli	taubench_airline_taubench_toolca	0.90	1.07	14.16	kept
taubench_airli	taubench_airline_taubench_toolca	0.83	1.08	6.38	kept
taubench_airli	taubench_airline_taubench_toolca	0.31	1.12	1.42	kept
taubench_airli	taubench_airline_taubench_toolca	8.18	1.10	3.84	kept
taubench_airli	taubench_airline_taubench_toolca	18.17	2.00	9.08	kept
taubench_airli	taubench_airline_taubench_toolca	32.24	2.00	4.52	kept
taubench_airli	taubench_airline_taubench_toolca	8.38	2.00	4.19	kept
taubench_airli	taubench_airline_taubench_toolca	4.51	2.00	2.26	kept
usaco	usaco_usaco_episodic__semantic_c	38.70	1.00	38.70	kept
usaco	usaco_usaco_episodic__semantic_c	276.19	2.00	138.09	kept
usaco	usaco_usaco_episodic__semantic_c	267.72	2.00	133.86	kept
usaco	usaco_usaco_episodic__semantic_d	80.04	2.00	40.02	kept
usaco	usaco_usaco_episodic__semantic_d	12.08	2.00	6.04	kept
usaco	usaco_usaco_episodic__semantic_g	28.10	2.00	14.05	kept
usaco	usaco_usaco_episodic__semantic_g	64.13	2.00	32.07	kept
usaco	usaco_usaco_episodic__semantic_o	286.50	2.00	28.65	kept
usaco	usaco_usaco_episodic__semantic_o	61.86	2.00	30.93	kept
usaco	usaco_usaco_episodic__semantic_o	22.49	2.00	11.25	kept
usaco	usaco_usaco_episodic__semantic_o	44.04	2.00	22.02	kept
usaco	usaco_usaco_episodic__semantic_o	21.14	2.00	10.57	kept

S10. The eight rerun pairs

Runs of one configuration (same agent, model and settings) on the same tasks, the data behind the run-to-run noise paragraph of the main text.

Table S28: The eight pairs of runs of one configuration. Acc. A, Acc. B: accuracy of the two runs, in per cent; Diff.: A minus B in points; Agree: share of tasks on which the two runs have the same outcome; Pred. sd: standard deviation of the difference in points if every task flipped independently at the observed rate of disagreement; z: the difference over that standard deviation. The sum of the squared z is the chi-square statistic of the main text.

Benchmark	Configuration	Tasks	Acc. A	Acc. B	Diff.	Agree	Pred. sd	z	Cost ratio
CORE-Bench hard	HAL Generalist Agent	45	8.9	22.2	-13.3	82	6.3	-2.1	1.72
GAIA	HAL Generalist Agent (claude-sonnet-4.5)	165	74.5	74.5	+0.0	82	3.3	+0.0	1.11
SWE-bench Verified mini	SWE-Agent(claude-3-7-sonnet-20250219)	50	52.0	54.0	-2.0	86	5.3	-0.4	1.02
SWE-bench Verified mini	SWE-Agent(o4-mini-2025-04-16)	50	56.0	52.0	+4.0	88	4.9	+0.8	1.05
SWE-bench Verified mini	SWE-Agent(claude-opus-4-1-20250805)	50	66.0	54.0	+12.0	72	7.5	+1.6	1.07
SWE-bench Verified mini	SWE-Agent(claude-opus-4-1-20250805)	50	66.0	68.0	-2.0	86	5.3	-0.4	1.30
SWE-bench Verified mini	SWE-Agent(claude-opus-4-1-20250805)	50	54.0	68.0	-14.0	78	6.6	-2.1	1.21
SWE-bench Verified mini	SWE-Agent(claude-sonnet-4-5-20250929)	50	72.0	68.0	+4.0	92	4.0	+1.0	1.04

S11. The frontier set against six alternatives

S12. The tie-robust constructions on known truth

The standard set and three tie-robust constructions in the scenarios of the main text, and the least favourable check for the gated set.

Table S30: The standard set and three tie-robust constructions on known truth, 300 boards per scenario. Kept: share of boards on which every truly undominated system is kept; size: mean number of systems kept; True: size of the true frontier. Gated: the gated step-down of Section 3.3; Step-down: the same without the gate, the version proved with no condition on the dependence of cost and accuracy errors; Interval: the interval construction of honestboard 0.5.1.

Scenario	K	T	True	Std kept	size	Gated kept	size	Step-down kept	size	Interval kept	size
Base	8	50	4.0	0.98	7.1	1.00	7.6	1.00	7.6	1.00	7.8
Base	8	150	3.9	0.97	6.2	1.00	6.7	1.00	6.8	1.00	7.0
Base	8	500	3.9	0.97	5.2	1.00	5.6	1.00	5.7	1.00	5.9
Base	20	50	5.8	0.98	16.9	1.00	18.5	1.00	18.8	1.00	19.1
Base	20	150	5.6	0.95	13.4	1.00	15.2	1.00	15.5	1.00	16.1

Scenario	K	T	True	Std kept	Std size	Gated kept	Gated size	Step-down kept	Step-down size	Interval kept	Interval size
Base	20	500	5.7	0.96	9.9	1.00	11.2	1.00	11.3	1.00	11.9
Fixed costs	20	150	5.9	1.00	13.6	1.00	13.5	1.00	13.8	1.00	14.4
Weak cost link	20	150	4.6	0.96	10.5	1.00	12.2	1.00	12.5	1.00	13.2
Crowded board	60	45	8.3	0.99	49.0	1.00	55.8	1.00	56.4	1.00	57.4
Low accuracy	20	65	5.9	0.98	18.0	1.00	19.2	1.00	19.3	1.00	19.5
Near-copies	20	150	4.9	0.98	13.1	1.00	15.4	1.00	15.8	1.00	16.4
Cost-outcome dep.	20	150	5.3	0.94	12.4	1.00	14.5	1.00	14.8	1.00	15.5
Near-tied costs	20	150	5.7	0.26	11.1	1.00	20.0	1.00	20.0	1.00	20.0
Clustered tasks	20	200	5.6	0.95	12.7	1.00	14.4	1.00	14.7	1.00	15.4
Cheaper near-copies	20	150	6.4	0.83	13.4	1.00	15.5	1.00	15.8	1.00	16.5

Table S31: Least favourable check: every system has the same true accuracy, so every system is in F and any exclusion is an error; true costs are spread so that their gaps are of the order of the critical value (cost spread, the standard deviation of the log true cost); failed tasks cost the failed-task multiple times as much as passed ones. Columns: share of boards on which the set keeps every system, at the nominal 0.95.

Cost spread	Failed-task multiple	K	T	Boards	Standard	Interval	Step-down	Gated
0.1	3	10	150	300	0.933	1.000	0.990	0.983
0.2	1	10	150	300	0.947	0.997	0.987	0.980
0.2	3	10	150	300	0.933	0.987	0.977	0.977
0.4	3	10	150	300	0.940	0.980	0.977	0.970

S13. The second leaderboard: prices, left-out models and frontiers

Table S32: The price table of the HELM audit: list prices in dollars per million tokens, with the source key of the price. Models from OpenAI and Anthropic that the HAL audit also prices use the HAL table; the others use a public price list fetched on 14 October 2026.

Model	Input ($/M)	Output ($/M)	Source
amazon__nova-lite-v1:0	0.06	0.24	LiteLLM list: amazon.nova-lite-v1:0
amazon__nova-micro-v1:0	0.035	0.14	LiteLLM list: amazon.nova-micro-v1:0
amazon__nova-premier-v1:0	2.5	12.5	LiteLLM list: amazon.nova-premier-v1:0
amazon__nova-pro-v1:0	0.8	3.2	LiteLLM list: amazon.nova-pro-v1:0
anthropic__claude-3-5-haiku-20241022	1	5	HAL price table: claude-3-5-haiku-20241022
anthropic__claude-3-5-sonnet-20241022	3	15	HAL price table: claude-3-5-sonnet-20241022
anthropic__claude-3-7-sonnet-20250219	3	15	HAL price table: claude-3-7-sonnet-20250219
anthropic__claude-haiku-4-5-20251001	1	5	LiteLLM list: claude-haiku-4-5-20251001
anthropic__claude-opus-4-20250514	15	75	HAL price table: claude-opus-4-20250514
anthropic__claude-sonnet-4-20250514	3	15	HAL price table: claude-sonnet-4-20250514
anthropic__claude-sonnet-4-5-20250929	3	15	HAL price table: claude-sonnet-4-5-20250929
meta__llama-3.1-405b-instruct-turbo	3.5	3.5	LiteLLM list: together_ai/meta-llama/Llama-3.1-405B-Instruct
meta__llama-3.1-70b-instruct-turbo	0.88	0.88	LiteLLM list: together_ai/meta-llama/Meta-Llama-3.1-70B-Instruct-Turbo
meta__llama-3.1-8b-instruct-turbo	0.18	0.18	LiteLLM list: together_ai/meta-llama/Meta-Llama-3.1-8B-Instruct-Turbo
meta__llama-4-scout-17b-16e-instruct	0.18	0.59	LiteLLM list: together_ai/meta-llama/Llama-4-Scout-17B-16E-Instruct
mistralai__mistral-7b-instruct-v0.3	0.2	0.2	LiteLLM list: together_ai/mistralai/Mistral-7B-Instruct-v0.3
mistralai__mixtral-8x7b-instruct-v0.1	0.6	0.6	LiteLLM list: together_ai/mistralai/Mixtral-8x7B-Instruct-v0.1
moonshotai__kimi-k2-instruct	1	3	LiteLLM list: together_ai/moonshotai/Kimi-K2-Instruct
openai__gpt-4.1-2025-04-14	2	8	HAL price table: gpt-4.1-2025-04-14
openai__gpt-4.1-mini-2025-04-14	0.4	1.6	HAL price table: gpt-4.1-mini-2025-04-14
openai__gpt-4.1-nano-2025-04-14	0.1	0.4	HAL price table: gpt-4.1-nano-2025-04-14
openai__gpt-4o-2024-11-20	2.5	10	HAL price table: gpt-4o-2024-11-20
openai__gpt-4o-mini-2024-07-18	0.15	0.6	HAL price table: gpt-4o-mini-2024-07-18
openai__gpt-oss-120b	0.15	0.6	LiteLLM list: together_ai/openai/gpt-oss-120b
openai__gpt-oss-20b	0.05	0.2	LiteLLM list: together_ai/openai/gpt-oss-20b
qwen__qwen2.5-72b-instruct-turbo	1.2	1.2	LiteLLM list: together_ai/Qwen/Qwen2.5-72B-Instruct-Turbo
qwen__qwen2.5-7b-instruct-turbo	0.3	0.3	LiteLLM list: together_ai/Qwen/Qwen2.5-7B-Instruct-Turbo
zai-org__glm-4.5-air-fp8	0.2	1.1	LiteLLM list: together_ai/zai-org/GLM-4.5-Air-FP8

Set	$K = 20$, $T = 50$ kept, all	size	$K = 20$, $T = 150$ kept, all	size	$K = 20$, $T = 500$ kept, all	size	$K = 60$, $T = 45$ kept, all	size	$K = 20$, $T = 65$ kept, all	size
Confidence set (joint margin)	0.98	16.9	0.95	13.4	0.96	9.9	0.99	49.0	0.98	18.0
Largest per-system margin	0.95	15.0	0.94	11.5	0.95	8.9	0.98	40.6	0.98	16.6
Per-system margins	0.92	14.1	0.92	10.8	0.93	8.5	0.90	36.5	0.95	15.8
Exact, per system	0.94	15.6	0.93	11.5	0.94	8.8	0.95	44.4	0.98	17.0
Exact, joint	0.99	18.1	0.96	14.1	0.97	10.2	0.99	55.4	1.00	18.9
Unadjusted pairwise tests	0.86	12.3	0.86	9.4	0.91	7.6	0.77	27.4	0.90	13.6
Frequency $\geq \alpha$	0.75	10.2	0.92	8.9	0.94	7.6	0.46	17.6	0.72	10.4
True frontier		5.8		5.6		5.7		8.3		5.9

Table S29: Seven ways to turn the same board into a set, on known truth (300 boards per scenario, per-task costs). Kept, all: share of boards on which every truly undominated system is kept (the joint guarantee; target 0.95). Size: mean number of systems kept. The confidence set of Section 3.3 of the main text is the first row; the next two are its per-system variants; the exact rows use one-sided paired tests with Bonferroni over each system's rivals or over every rival pair; "unadjusted" keeps a system unless some cheaper rival beats it at the 5% level with no correction; "frequency" keeps a system that is on the observed frontier in at least 5% of the task resamples.

Table S33: HELM Capabilities models left out of every board, with the reason. WildBench is left out of the audit altogether because it records no prompt tokens.

Model	Why it is left out
deepseek-ai__deepseek-r1-0528	completion tokens are not recorded
deepseek-ai__deepseek-v3	completion tokens are not recorded
google__gemini-1.5-flash-002	completion tokens are not recorded
google__gemini-1.5-pro-002	completion tokens are not recorded
google__gemini-2.0-flash-001	completion tokens are not recorded
google__gemini-2.0-flash-lite-preview-02-05	completion tokens are not recorded
google__gemini-2.5-flash-lite	completion tokens are not recorded
google__gemini-2.5-flash-preview-04-17	completion tokens are not recorded
google__gemini-2.5-pro-preview-03-25	completion tokens are not recorded
google__gemini-3-pro-preview	completion tokens are not recorded
writer__palmyra-med	completion tokens are not recorded
writer__palmyra-x5	completion tokens are not recorded
anthropic__claude-opus-4-20250514-thinking-10k	hidden reasoning tokens are not recorded
anthropic__claude-sonnet-4-20250514-thinking-10k	hidden reasoning tokens are not recorded
openai__gpt-5-2025-08-07	hidden reasoning tokens are not recorded
openai__gpt-5-mini-2025-08-07	hidden reasoning tokens are not recorded
openai__gpt-5-nano-2025-08-07	hidden reasoning tokens are not recorded
openai__gpt-5.1-2025-11-13	hidden reasoning tokens are not recorded
openai__o3-2025-04-16	hidden reasoning tokens are not recorded
openai__o4-mini-2025-04-16	hidden reasoning tokens are not recorded
qwen__qwen3-next-80b-a3b-thinking	hidden reasoning tokens are not recorded
xai__grok-3-mini-beta	hidden reasoning tokens are not recorded
xai__grok-4-0709	hidden reasoning tokens are not recorded
allenai__olmo-2-0325-32b-instruct	no price under the pricing rule
allenai__olmo-2-1124-13b-instruct	no price under the pricing rule
allenai__olmo-2-1124-7b-instruct	no price under the pricing rule
allenai__olmoe-1b-7b-0125-instruct	no price under the pricing rule
ibm__granite-3.3-8b-instruct	no price under the pricing rule
ibm__granite-4.0-h-small	no price under the pricing rule
ibm__granite-4.0-micro	no price under the pricing rule
marin-community__marin-8b-instruct	no price under the pricing rule
meta__llama-4-maverick-17b-128e-instruct-fp8	no price under the pricing rule
mistralai__mistral-large-2411	no price under the pricing rule
mistralai__mistral-small-2503	no price under the pricing rule
mistralai__mixtral-8x22b-instruct-v0.1	no price under the pricing rule
qwen__qwen3-235b-a22b-fp8-tput	no price under the pricing rule
qwen__qwen3-235b-a22b-instruct-2507-fp8	no price under the pricing rule
writer__palmyra-fin	no price under the pricing rule
writer__palmyra-x-004	no price under the pricing rule
xai__grok-3-beta	no price under the pricing rule

Table S34: Sensitivity of the HELM audit to the host of the open-weight prices: the models that LiteLLM also lists at DeepInfra are repriced there from the same recorded tokens. Frontier and set: sizes under the Together AI prices of the audit and at DeepInfra; moved: systems that enter or leave; cost ratio: mean ratio of the repriced models' cost per task.

Benchmark	Systems	Frontier	at DeepInfra	moved	Set	at DeepInfra	moved	Cost ratio
GPQA	28	6	6	4	17	14	5	0.46
IFEval	28	5	6	1	11	13	2	0.46
MMLU-Pro	28	6	5	5	13	13	4	0.45
Omni-MATH	28	6	3	3	10	5	5	0.47

S14. Fresh-task replication of the certificates, board by board

Table S35: Fresh-task replication of the entrant certificates, board by board. Cases: entrant-holder pairs tested over the random splits; distinct pairs: how many different entrant-holder pairs they are; leads in points. Groups as in Table 9 of the main text.

Leaderboard	Board	Group	Cases	Distinct pairs	Lead on D	Lead on V	Positive	Significant
HAL	ColBench backend programming	certified	14	7	9.4	5.1	0.86	0.43
HAL	ColBench backend programming	nominal only	49	8	5.9	4.6	1.00	0.59
HAL	ColBench backend programming	neither	212	17	2.9	3.0	0.86	0.40
HAL	ColBench frontend design	nominal only	1	1	3.0	-2.3	0.00	0.00
HAL	ColBench frontend design	neither	32	3	0.8	-0.2	0.53	0.00
HAL	GAIA	nominal only	6	6	16.3	4.8	0.50	0.33
HAL	GAIA	neither	337	56	5.2	6.2	0.79	0.21
HAL	Online-Mind2Web	certified	34	1	14.9	13.1	1.00	1.00
HAL	Online-Mind2Web	nominal only	5	1	11.7	16.3	1.00	1.00
HAL	Online-Mind2Web	neither	266	38	3.5	2.6	0.76	0.11
HAL	ScienceAgentBench	nominal only	3	3	18.3	1.3	0.67	0.00
HAL	ScienceAgentBench	neither	173	25	7.9	7.3	0.87	0.20
HAL	USACO	certified	48	4	13.8	9.2	0.98	0.83
HAL	USACO	nominal only	46	5	10.6	10.0	0.96	0.76
HAL	USACO	neither	90	6	4.4	4.8	0.83	0.43
HELM	GPQA	certified	88	13	14.1	14.0	0.99	0.92
HELM	GPQA	nominal only	52	12	10.5	10.6	0.98	0.90
HELM	GPQA	neither	189	19	4.7	6.0	0.77	0.49
HELM	IFEval	certified	67	7	7.2	6.2	1.00	0.87
HELM	IFEval	nominal only	29	6	5.8	7.1	0.97	0.97
HELM	IFEval	neither	58	4	2.5	3.5	0.90	0.43
HELM	MMLU-Pro	certified	148	10	13.5	14.0	1.00	0.97
HELM	MMLU-Pro	nominal only	93	9	4.6	4.4	1.00	0.62
HELM	MMLU-Pro	neither	116	8	2.9	4.0	1.00	0.85
HELM	Omni-MATH	certified	203	9	17.1	17.6	1.00	1.00
HELM	Omni-MATH	nominal only	3	2	4.7	0.6	0.33	0.33
HELM	Omni-MATH	neither	63	4	2.1	1.1	0.76	0.08

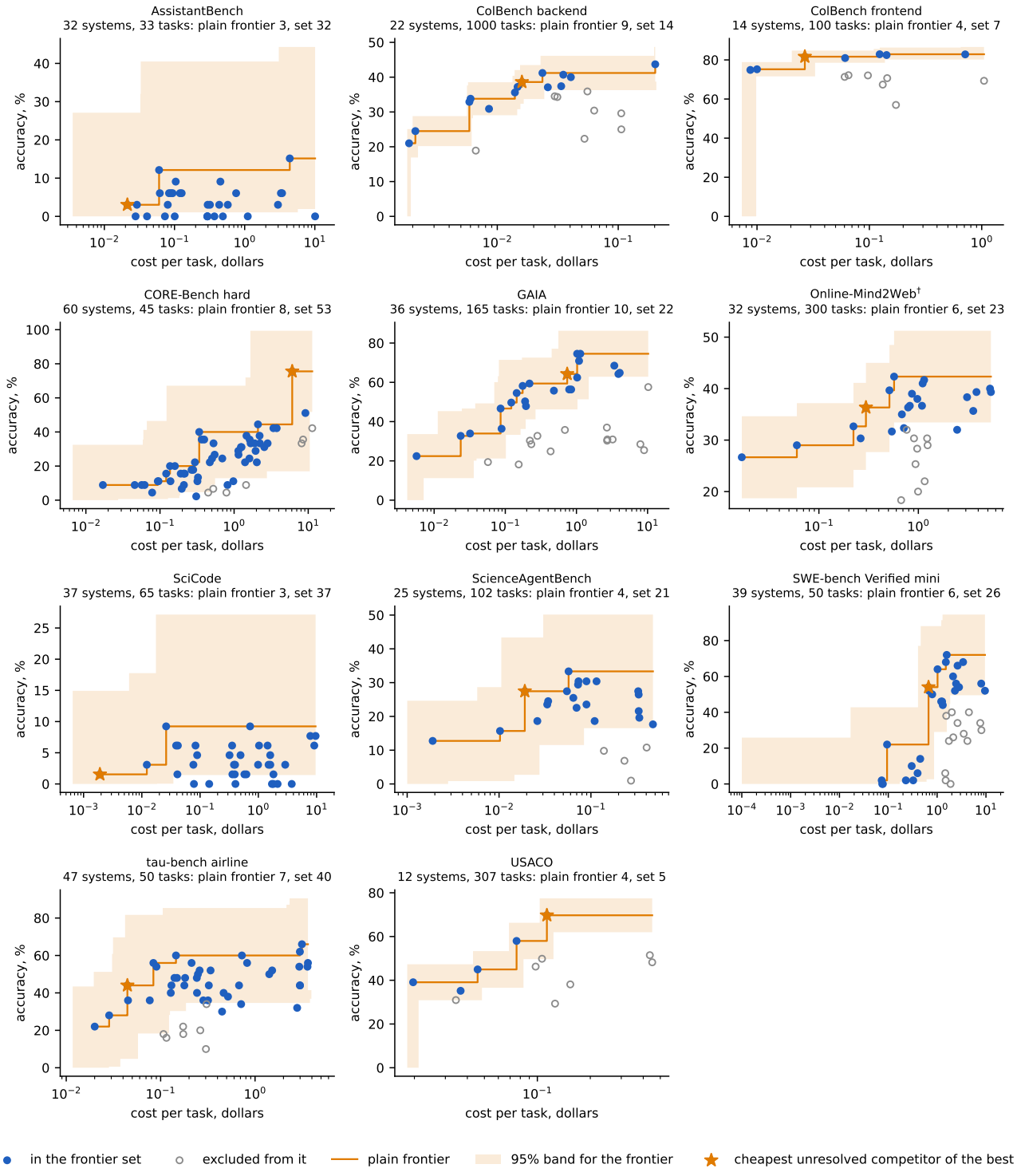


Figure 1: Cost per task against accuracy on every HAL benchmark. The orange staircase is the observed frontier; filled points are systems that cannot be excluded from the frontier at the 5% level; the star is the cheapest unresolved competitor of the best; the shaded region is the 95% simultaneous band for the frontier curve.

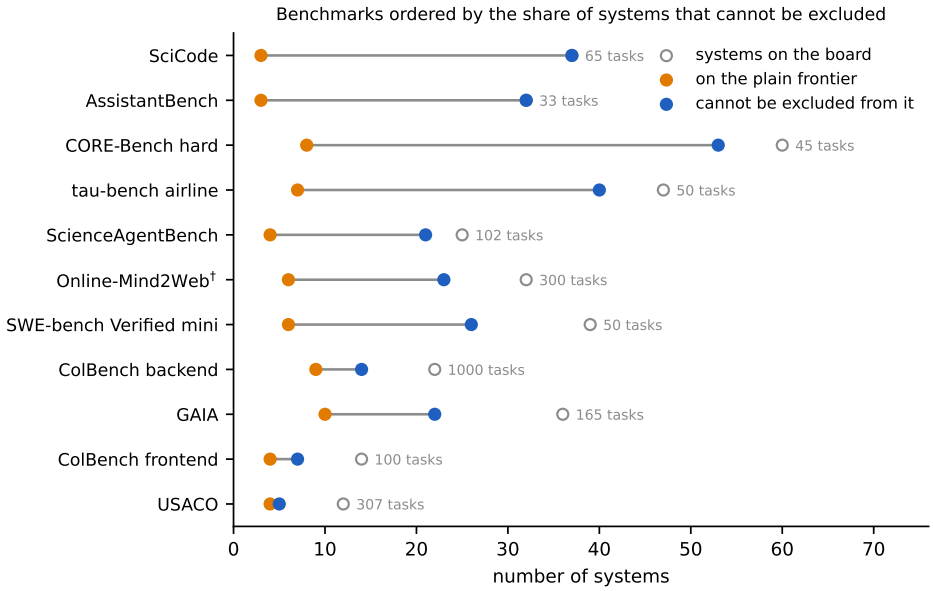


Figure 2: For each benchmark, the number of systems on the plain frontier (orange), the number that cannot be excluded from it at the 5% level (blue), and the number of systems on the board (hollow), with the task count beside it.

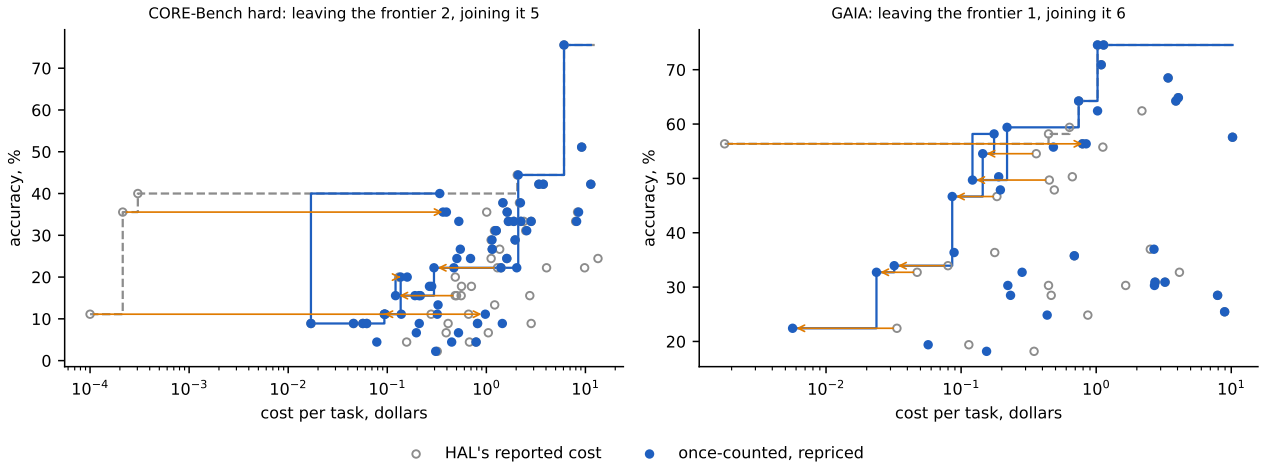


Figure 3: HAL's published frontier (hollow points, dashed staircase, HAL's reported costs) against the frontier under once-counted, repriced costs (filled points, solid staircase) on the two boards that change most. Arrows join the two costs of each system whose frontier status changes. Costs of zero are drawn at 10^{-4} dollars.

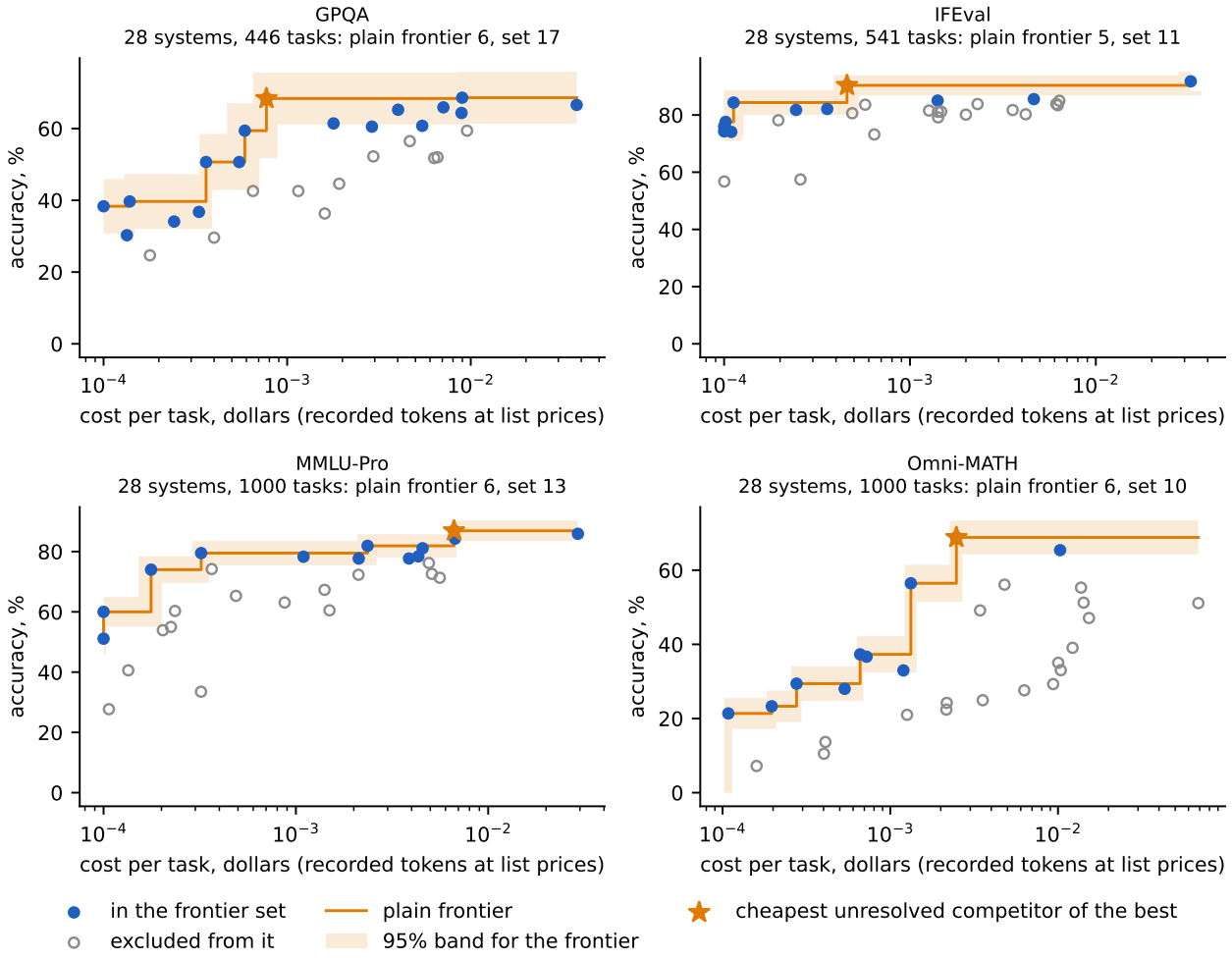


Figure 4: Cost per task against accuracy on the four HELM Capabilities boards. The orange staircase is the observed frontier; filled points are systems that cannot be excluded from the frontier at the 5% level; the star is the cheapest unresolved competitor of the best; the shaded region is the 95% simultaneous band. Costs below 10^{-4} dollars are drawn at 10^{-4}.